

# LLMs in 2025

## crash-course

Agent(MCP(tools(rag(llm)))) ?????

# Who am I?

- Jiaqi Chen
- 24 yr
- BSc AI, University Utrecht
- Start-ups / Entrepreneurship
  - Own startup
  - incubator board year
  - DialogueTrainer Product Manager
- Bouldering, live music, photography

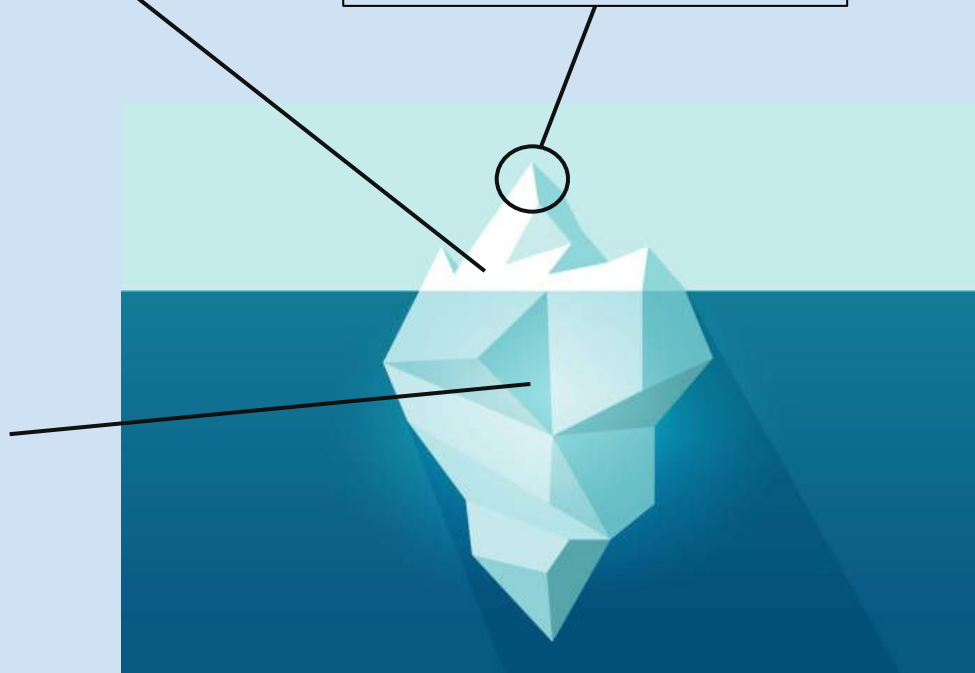


# Preface

# Techichal

Societal,  
mental/cognitive,  
political,  
Economic,  
Cultural,  
Academic,  
Environmental,  
Ethical,  
Educational

This presentation



"Crash course in LLMs"

Techichal

This presentation

Costs

Effects

Costs

Effects

Societal,  
Mental/cognitive  
political,  
Economic,  
Cultural,  
Academic,  
Environmental,  
Ethical,  
Educational

"Crash course in LLMs"

Costs Effects Costs Effects Costs

Technical

Costs Effects Costs Effects Costs

Societal,

Costs Effects Costs Effects Costs

Mental/cognitive

Costs Effects Costs Effects Costs

political,

Costs Effects Costs Effects Costs

Economic,

Costs Effects Costs Effects Costs

Cultural,

Costs Effects Costs Effects Costs

Academic,

Costs Effects Costs Effects Costs

Environmental,

Costs Effects Costs Effects Costs

Ethical,

Costs Effects Costs Effects Costs

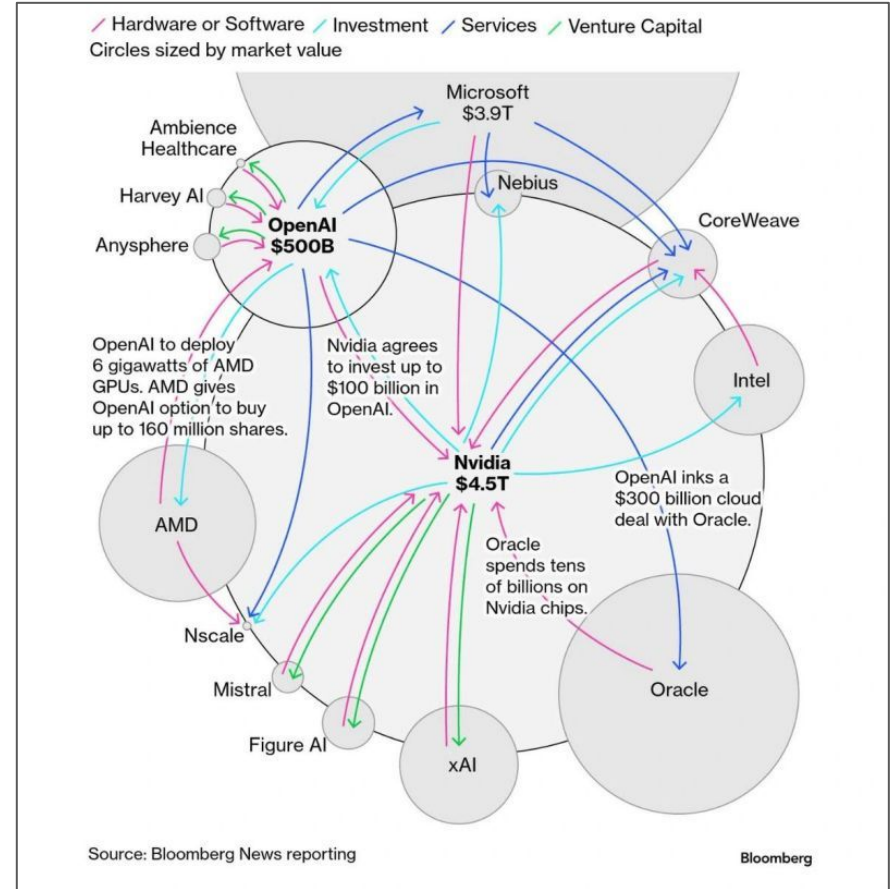
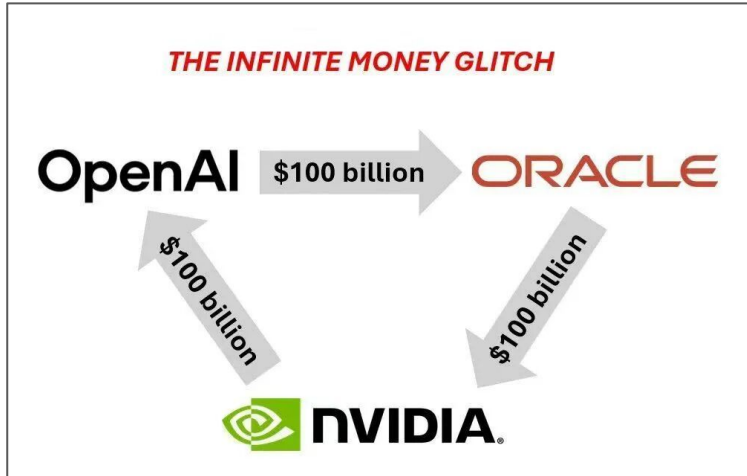
Educational

This presentation

Crash Course in LLMs

# Financial 'dimension'

The investment space of tech in 2025



# Human centric AI



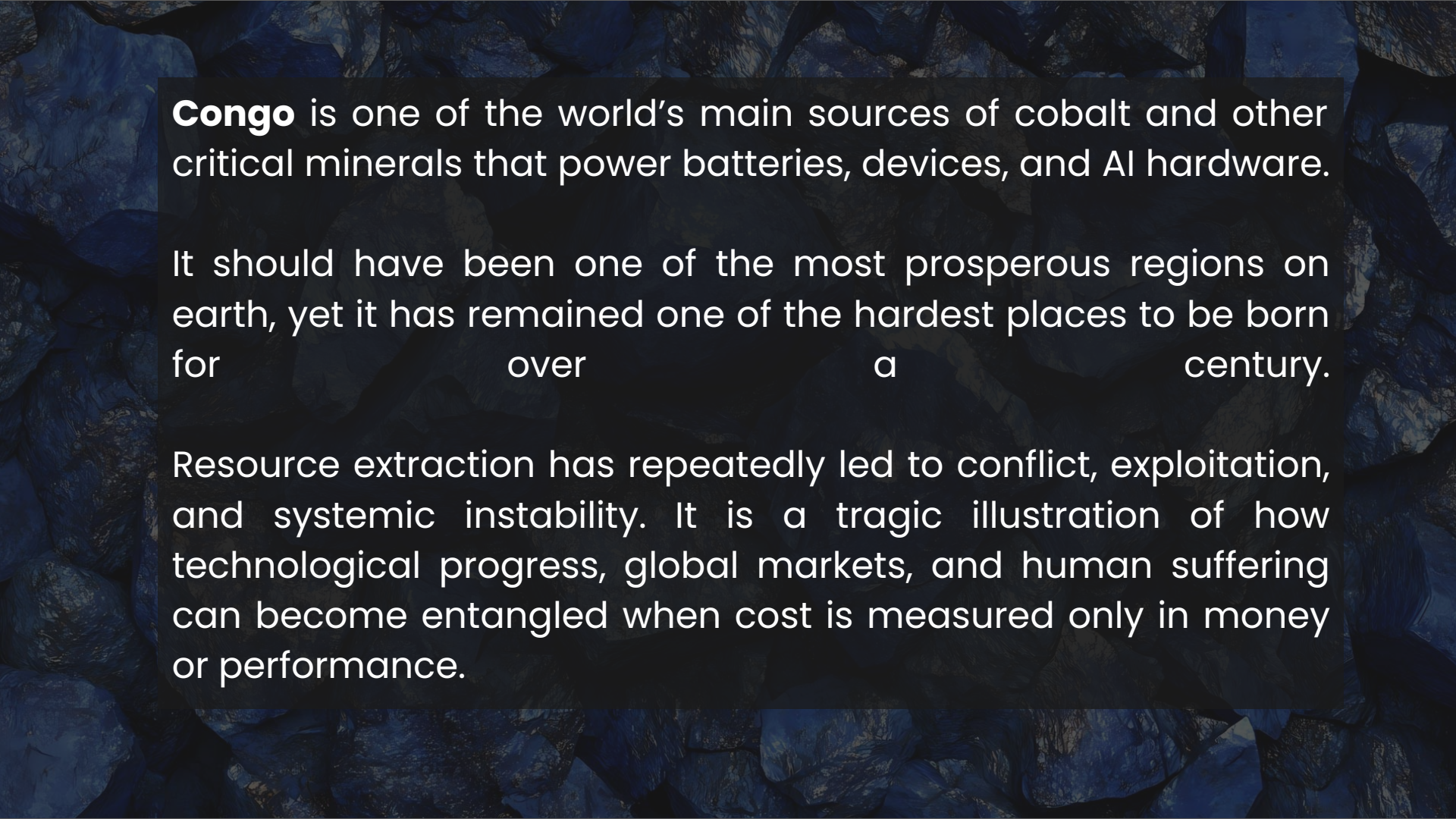
## ★ AI Overview

Tantalum is used primarily in electronics as a component for capacitors and resistors in devices like phones and computers. It is also used in high-temperature alloys for jet engines and nuclear reactors, and due to its corrosion resistance and biocompatibility, it is ideal for medical implants and surgical devices. 



## ★ AI Overview

Cobalt is used primarily in lithium-ion batteries for electronics and electric vehicles, in high-strength, heat-resistant alloys for jet engines and turbines, and in pigments to create vibrant blue and green colors. It is also used in magnetic materials, as a catalyst in the petroleum industry, and the radioactive isotope cobalt-60 is used for cancer treatment and as a gamma-ray source. 

The background of the entire image is a dark, textured surface resembling jagged rocks or mineral deposits in shades of deep blue and black. A semi-transparent dark grey rectangular box is centered on the image, containing white text.

**Congo** is one of the world's main sources of cobalt and other critical minerals that power batteries, devices, and AI hardware.

It should have been one of the most prosperous regions on earth, yet it has remained one of the hardest places to be born for over a century.

Resource extraction has repeatedly led to conflict, exploitation, and systemic instability. It is a tragic illustration of how technological progress, global markets, and human suffering can become entangled when cost is measured only in money or performance.

Arthur C. Clarke's Third Law:

Any sufficiently  
advanced  
technology is  
indistinguishable  
from magic

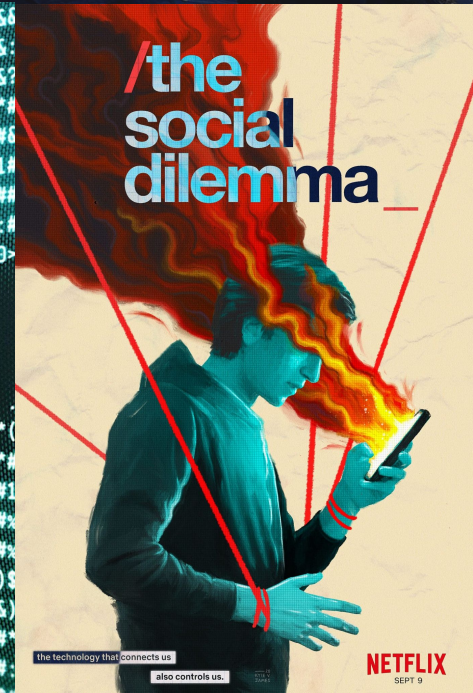
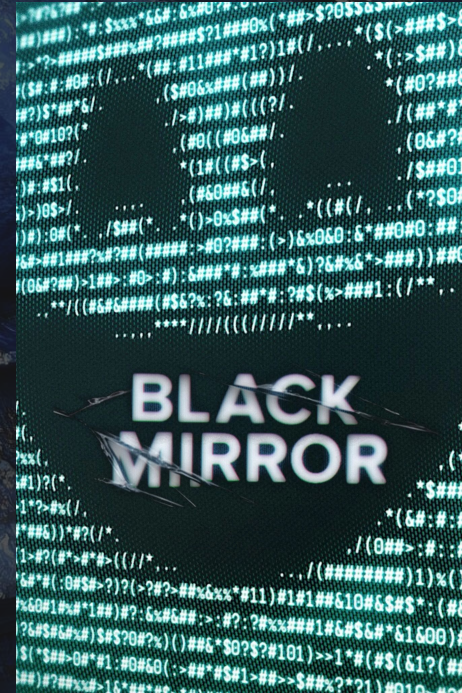
Techichal

Societal,  
mental/cognitive,  
political,  
Economic,  
Cultural,  
Academic,  
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Ethical,  
Educational

Arthur C. Clarke's Third Law:

Any sufficiently  
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Magic



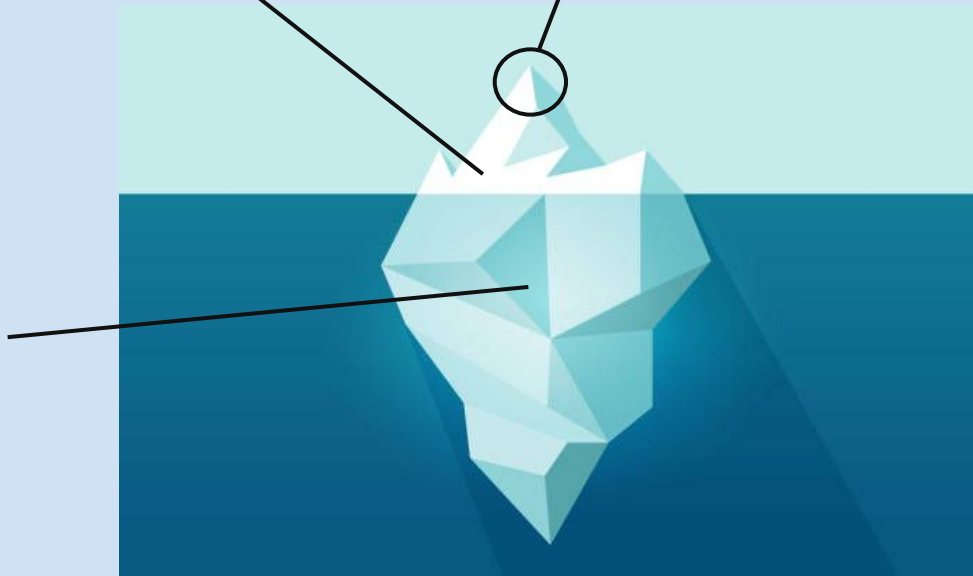
AI Projects

| AI projects      |          |         |
|------------------|----------|---------|
| Clients          | Students | Mentors |
| Bondfish         |          |         |
|                  |          |         |
|                  |          |         |
|                  |          |         |
| Gradient Metrics |          |         |
|                  |          |         |
|                  |          |         |
|                  |          |         |
| Profectus        |          |         |
|                  |          |         |
|                  |          |         |
|                  |          |         |

# Techichal

Societal,  
mental/cognitive,  
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Educational

This presentation



"Crash course in LLMs"

Peak of Inflated Expectations

Plateau of Productivity

AI, including LLMs,  
will change the world

a lot

Age of Enlightenment

Trough of Disillusionment

Innovation trigger

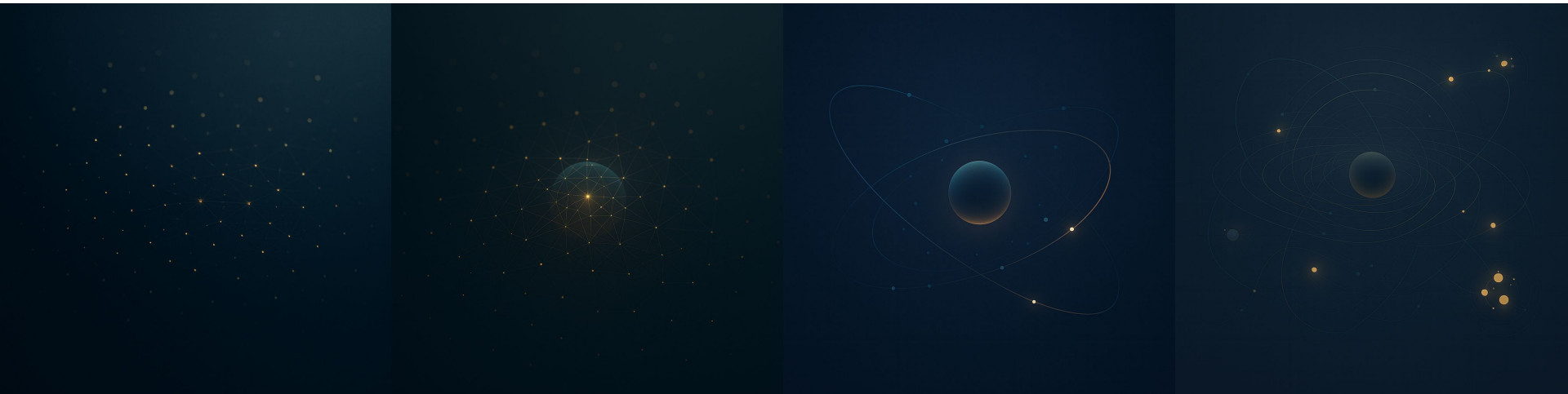
Innovation graveyard

Pre-LLM

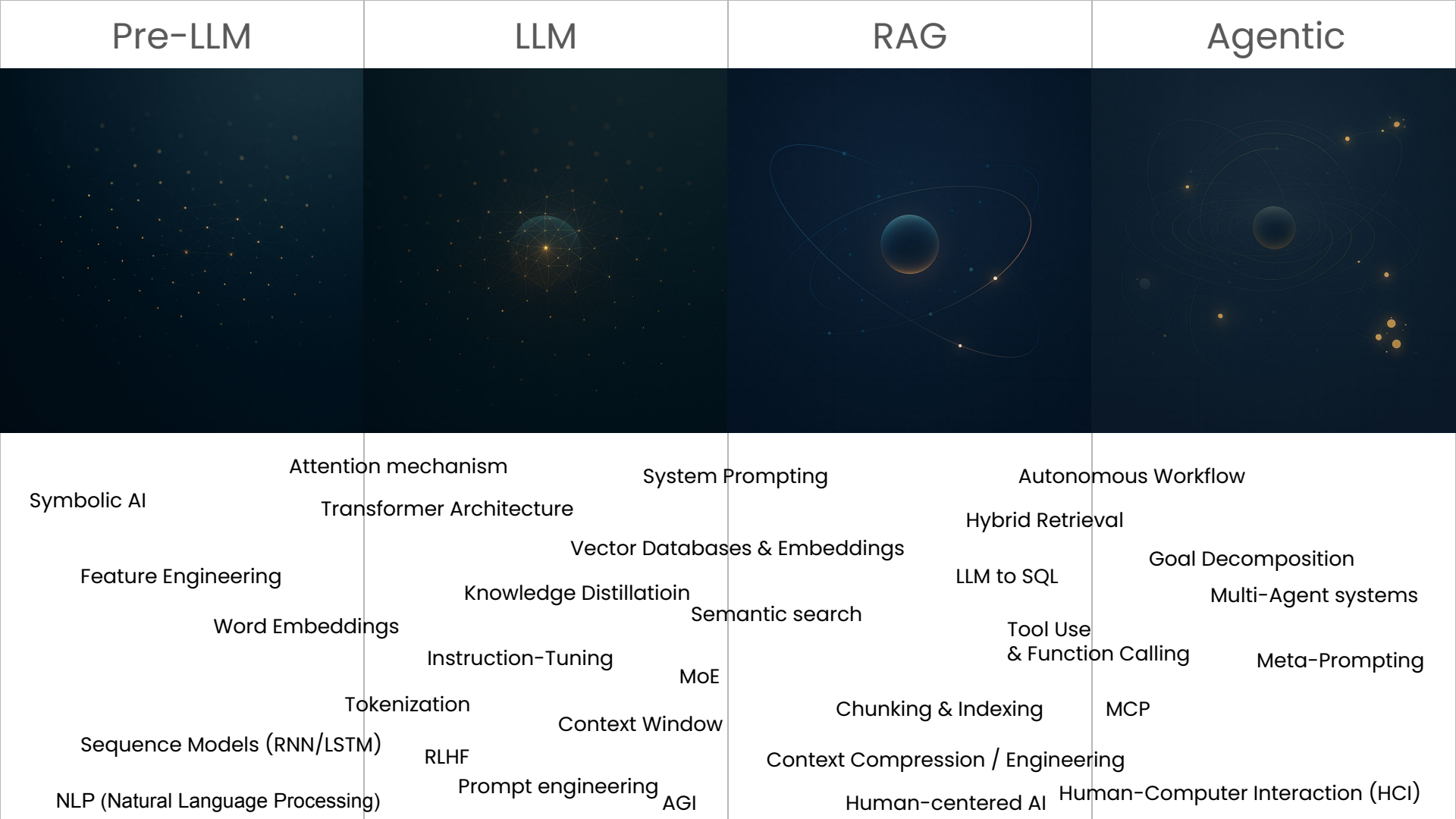
LLM

RAG

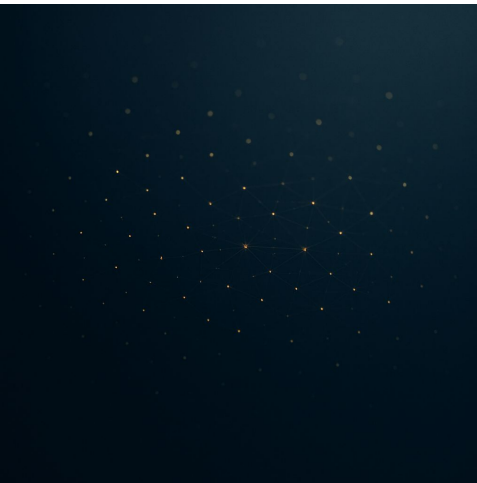
Agentic



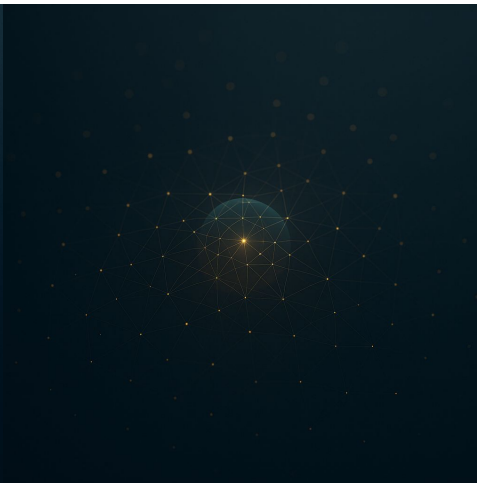
Terms don't strictly follow timeline



Pre-LLM



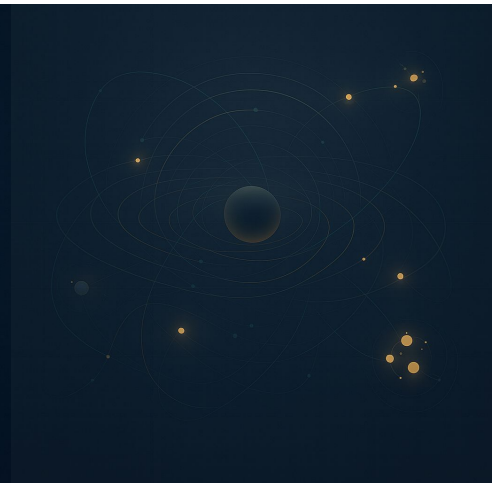
LLM



RAG



Agentic



Data pipelines,  
Sklearn pipelines  
Embeddings  
Tokenization

LLM/SLM  
LLM  
pipelines

Basic RAG  
(Hybrid)  
indexing

LLM  
interactions  
overview

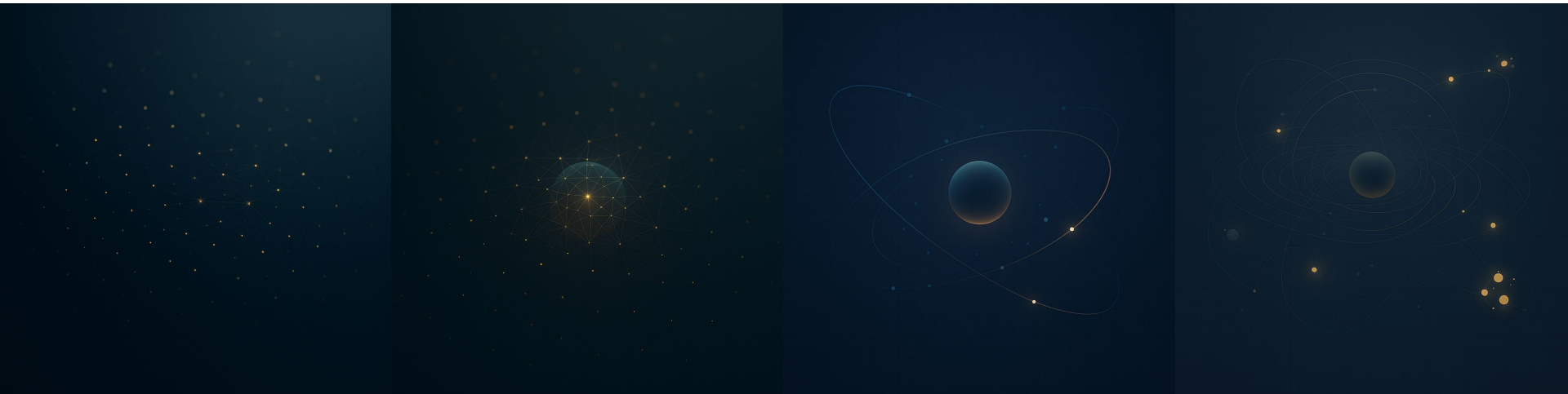
MCP

Pre-LLM

LLM

RAG

Agentic



Data-pipelines  
Model-pipelines

Vectors &  
embeddings

LLMs

(as a  
data-pipeline)

RAG

(as a  
LLM-pipeline)

Agent

(as an LLM with  
reasoning and  
tools)

Pre  
LLM

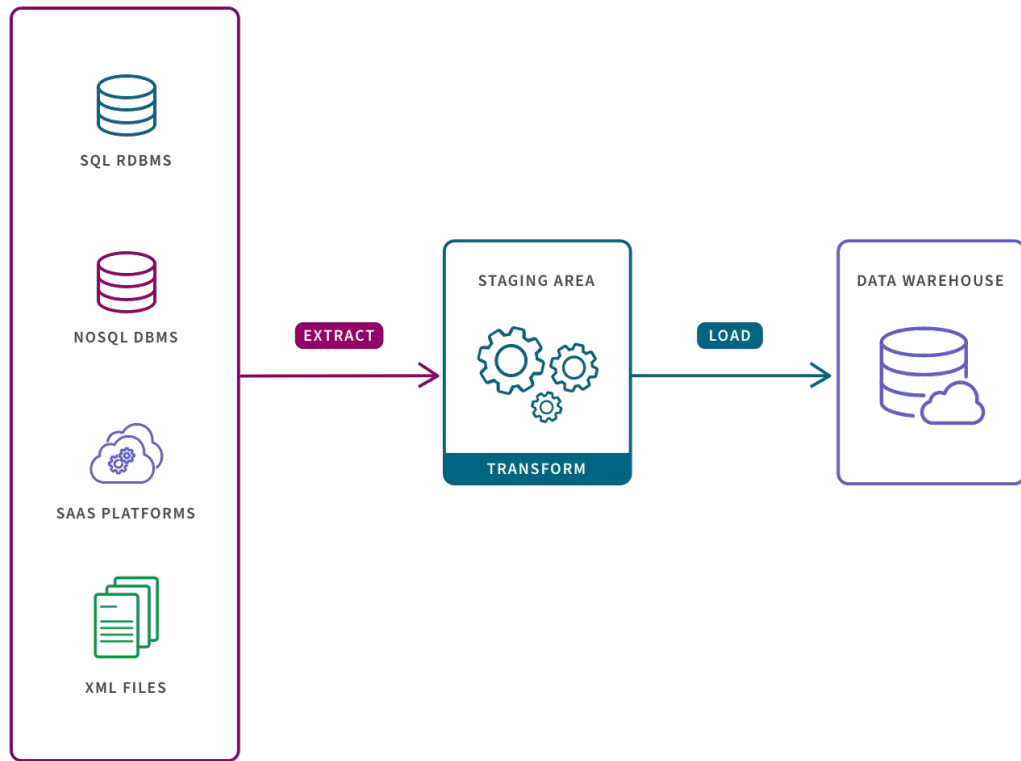


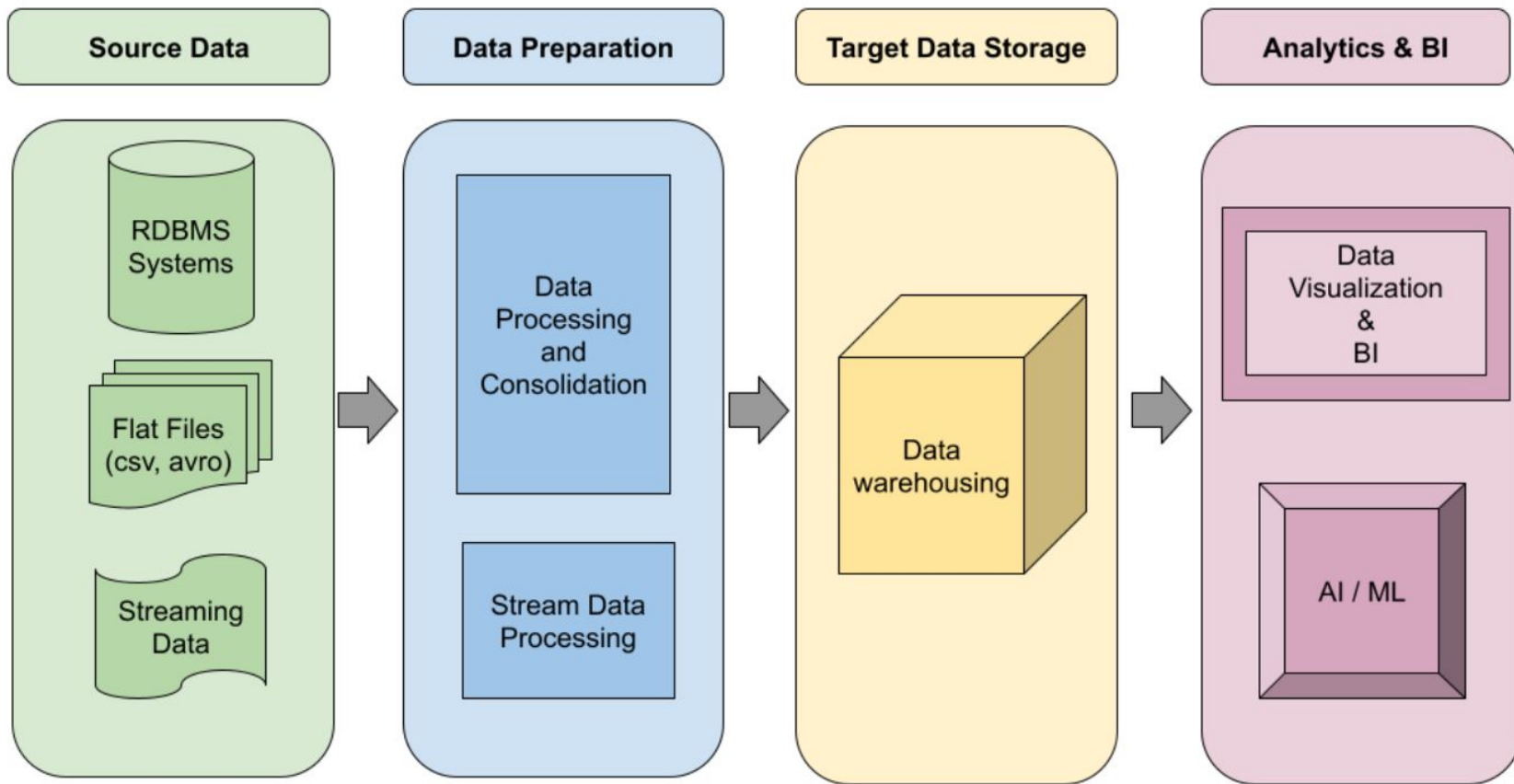
# Data Pipelines



Level 1: input-output

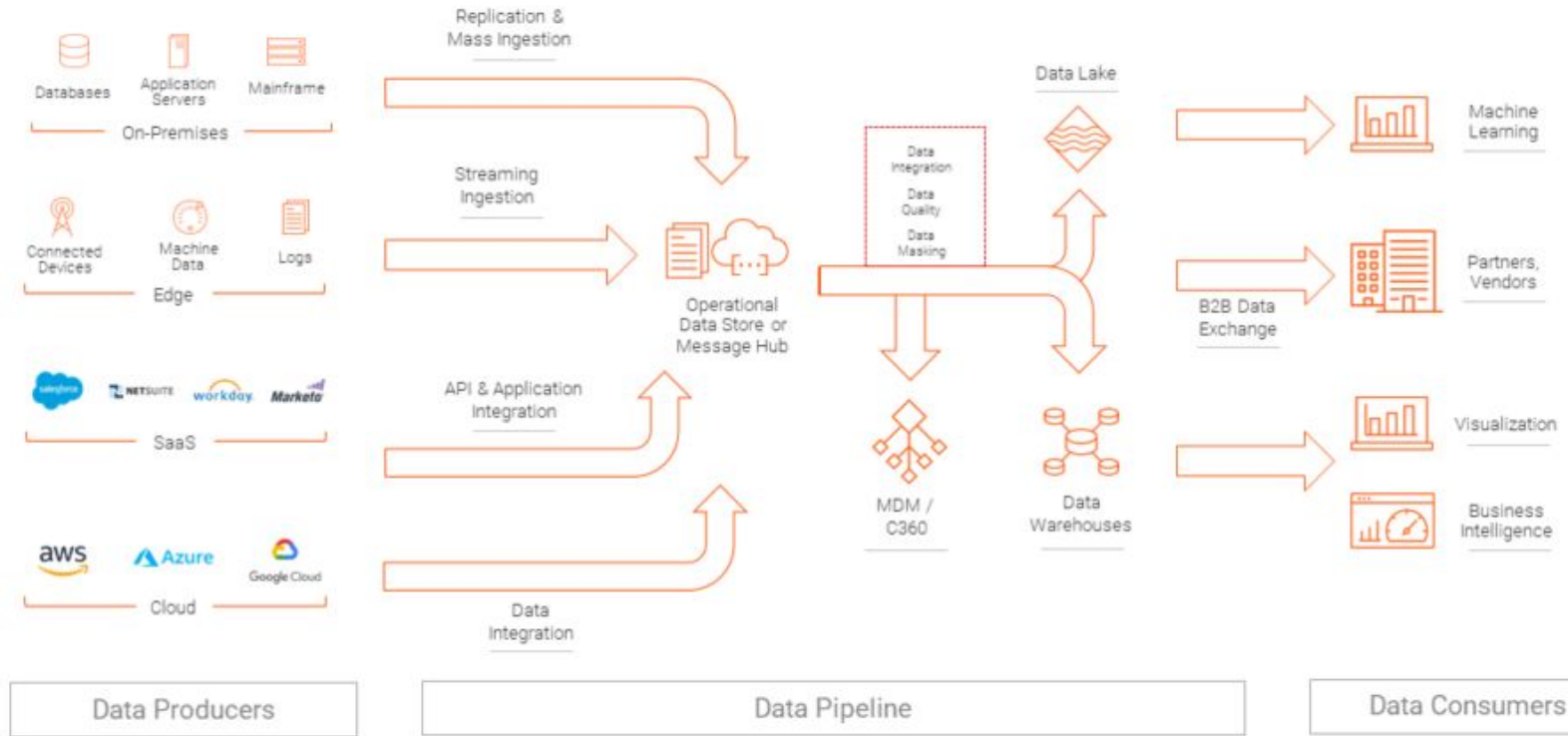
# ETL (Extract, Transform, Load)





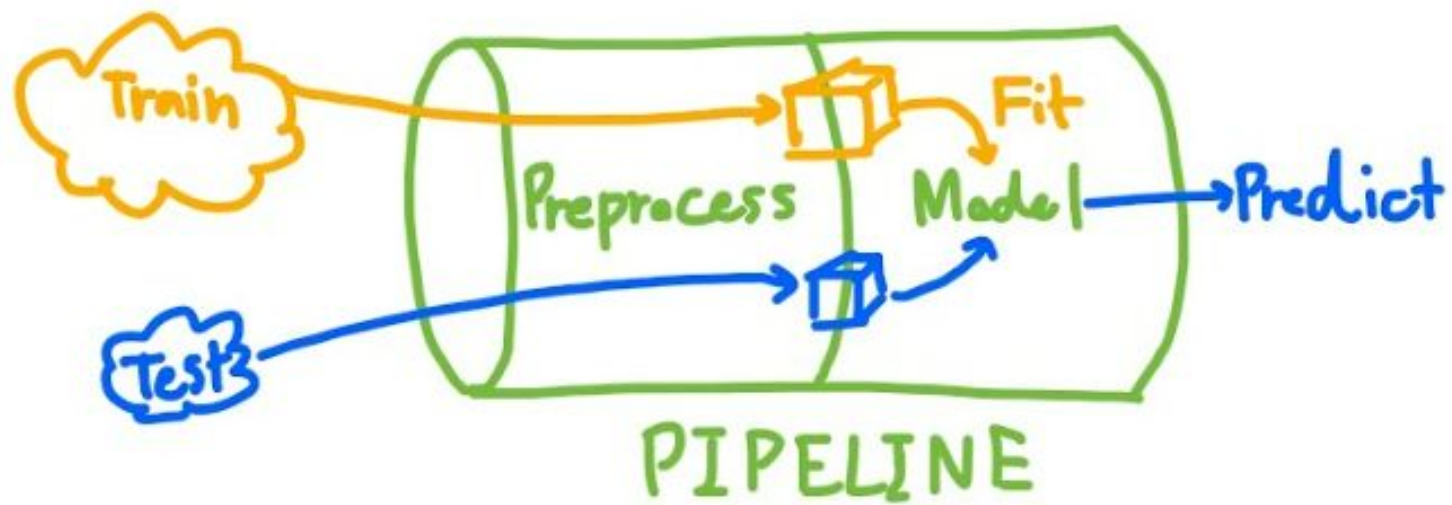
Level 3: high level end to end

# Data Pipeline Patterns

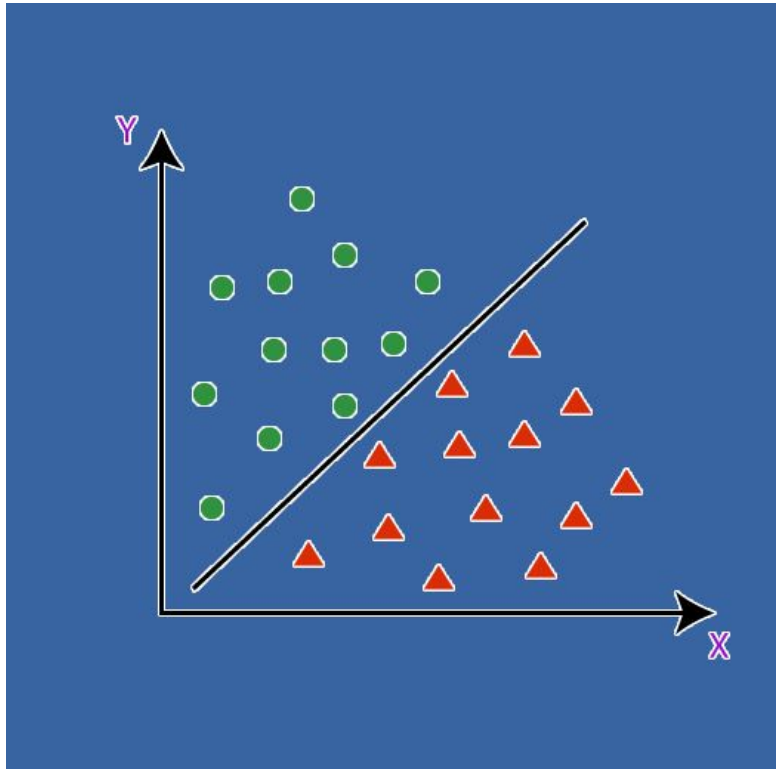


Level 4: integrations and more choices

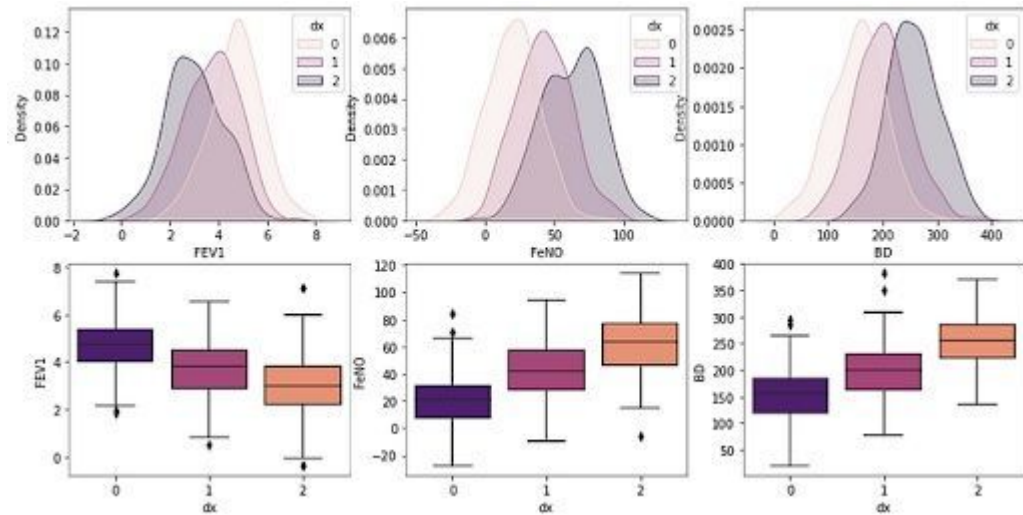
# Pre-llm models (AI pre-covid)



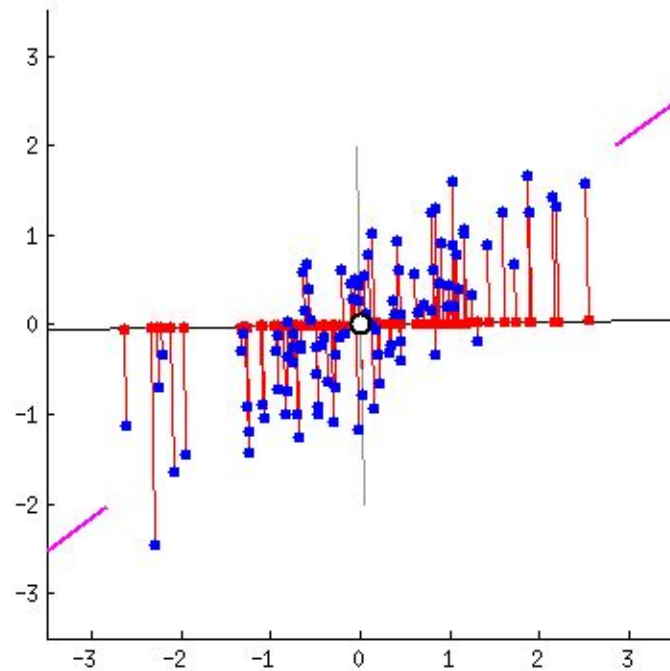
# SVM



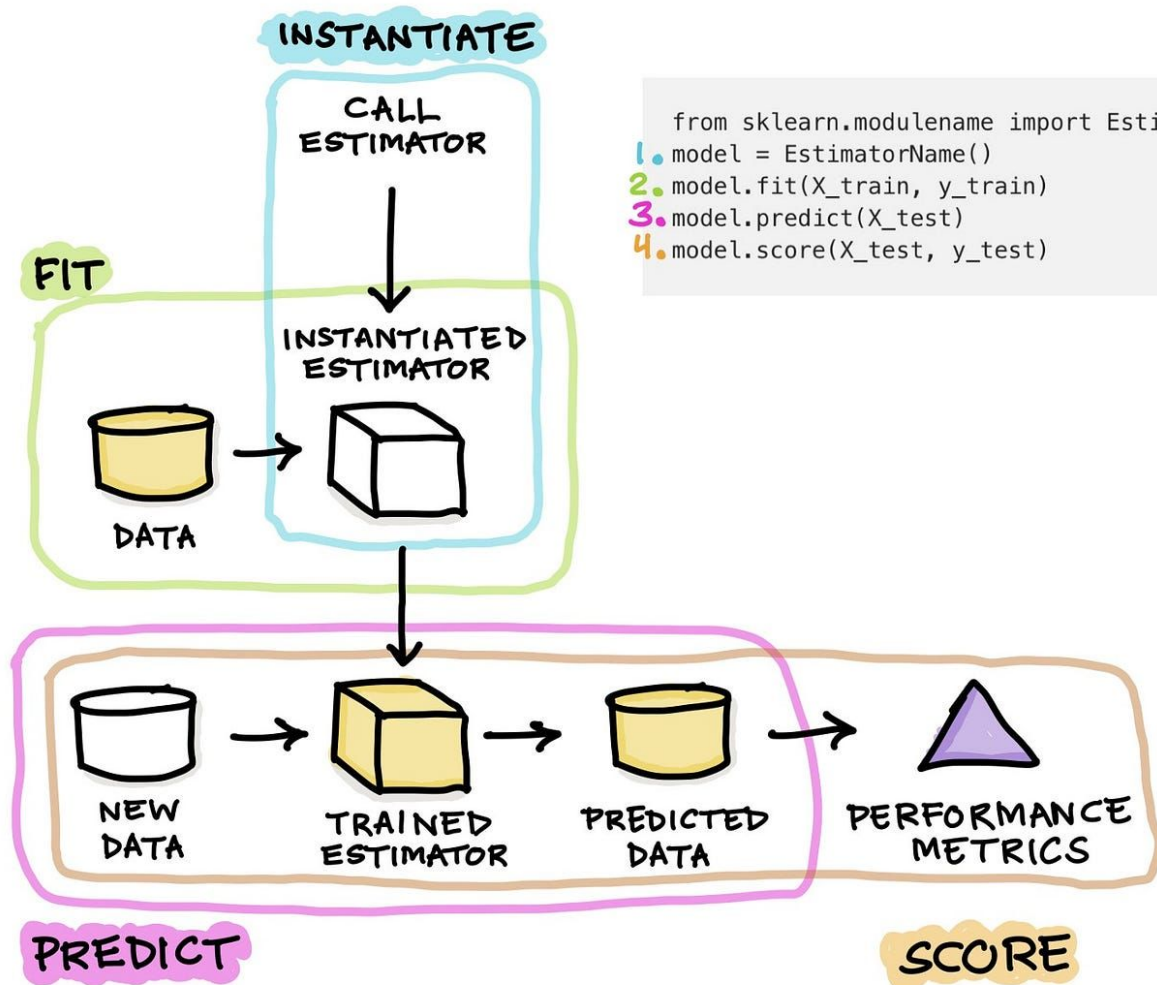
# Bayes



# PCA

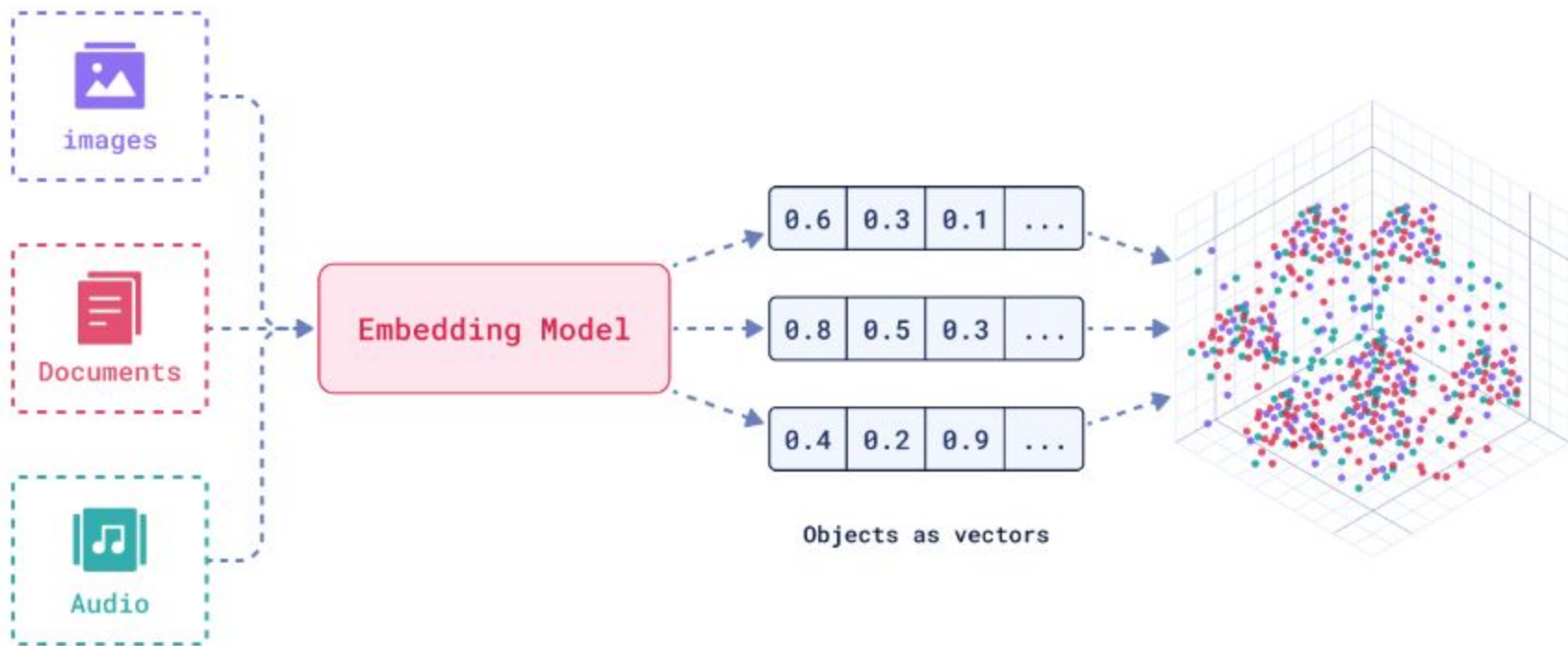


# Sk-Learn

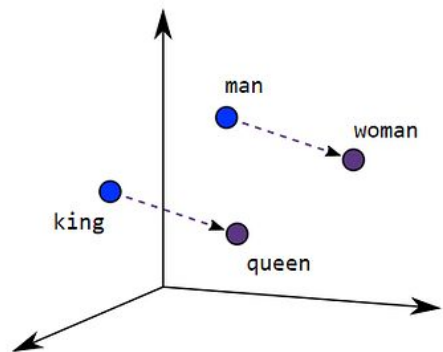


```
from sklearn.module_name import EstimatorName
1. model = EstimatorName()
2. model.fit(X_train, y_train)
3. model.predict(X_test)
4. model.score(X_test, y_test)
```

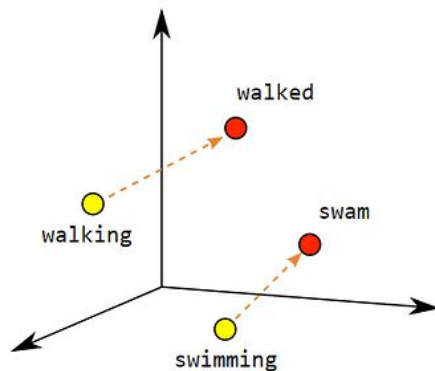
# Embeddings



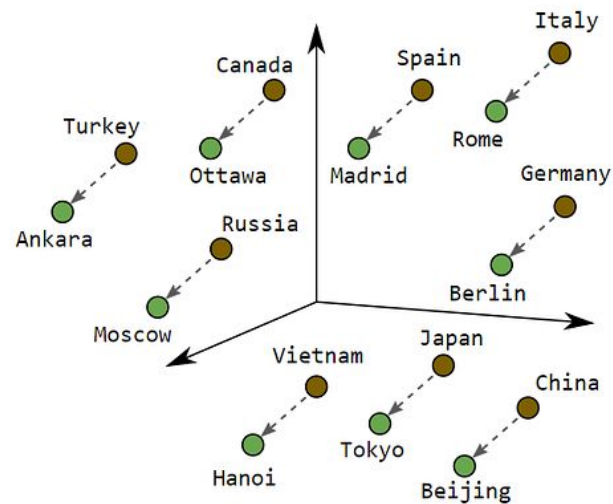
Embeddings can capture relations between the vectors (in a latent space)



Male-Female

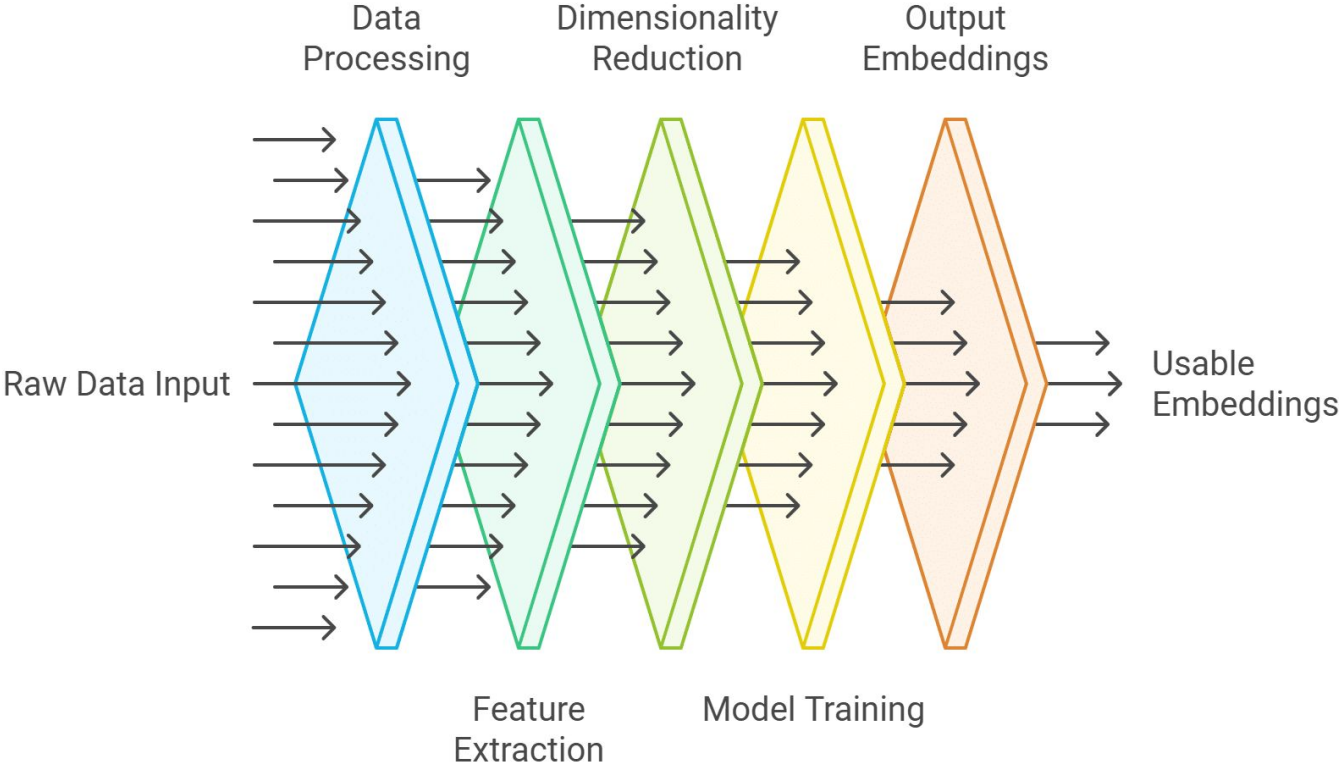


Verb Tense



Country-Capital

# Embedding Model Process



# Tokenization

# Tiktokenizer

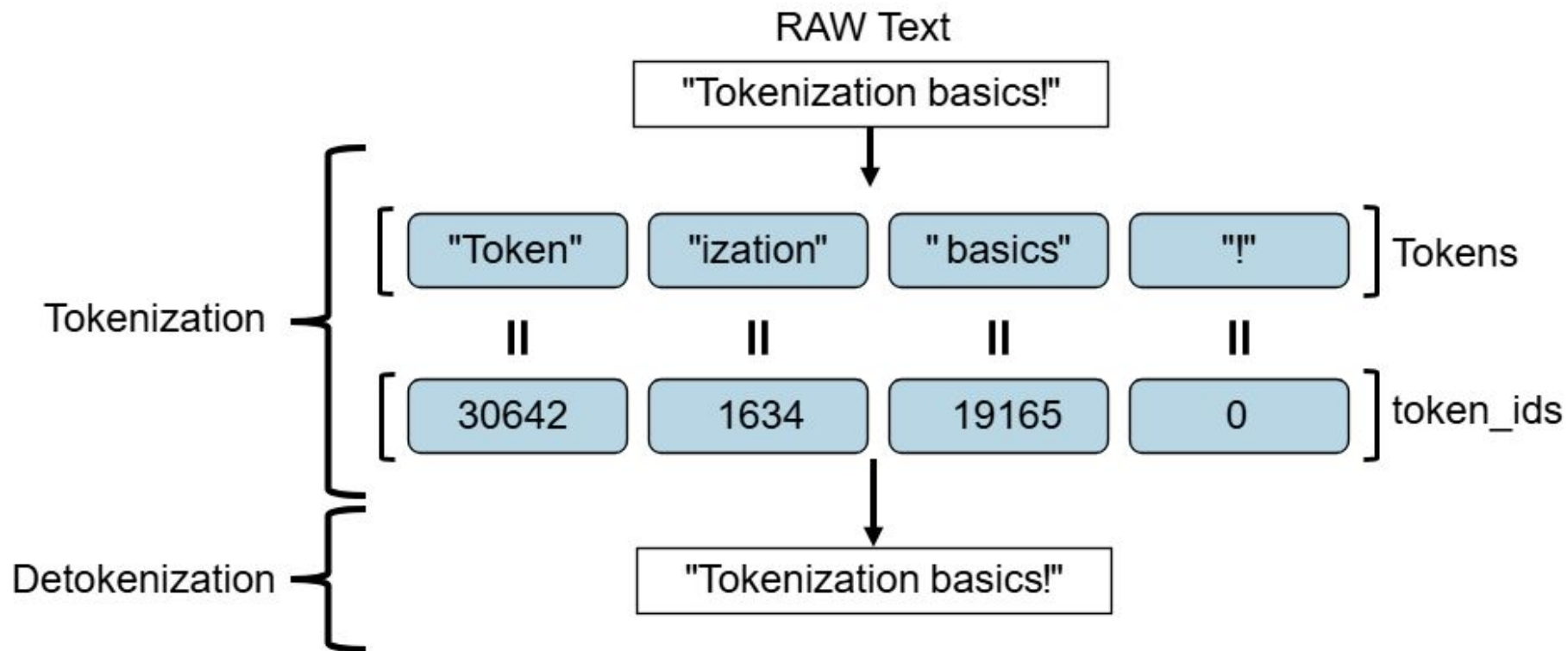
cl100k\_base

```
def fizz():  
    for i in range(1, 101):  
        if i % 5 == 0 and i % 3 == 0:  
            print("fizzbuzz")  
        elif i % 5 == 0:  
            print("buzz")  
        elif i % 3 == 0:  
            print("fizz")  
        else:  
            print(i)
```

Token count

77

```
def·fizz():\n···for·i·in·range(1,·101):\n·····if·i·%·5·==·0·and·i·%·3·==·0:\n······print("fizzbuzz")\n·····elif·i·%·5·==·0:\n······print("buzz")\n·····elif·i·%·3·==·0:\n······print("fizz")\n·····else:\n······print(i)
```



Token count  
77

```
def fizz():\n    for i in range(1, 101):\n        if i % 5 == 0 and i % 3 == 0:\n            print("fizzbuzz")\n        elif i % 5 == 0:\n            print("buzz")\n        elif i % 3 == 0:\n            print("fizz")\n        else:\n            print(i)
```

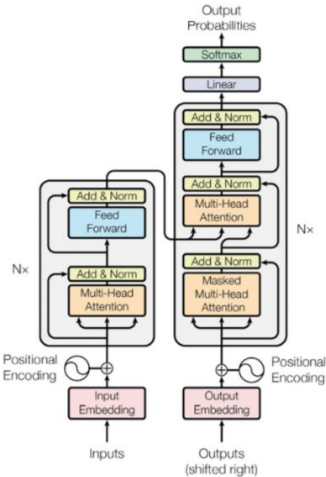
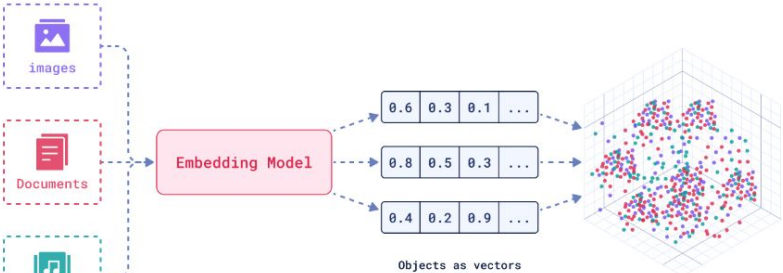
# Tiktokerizer

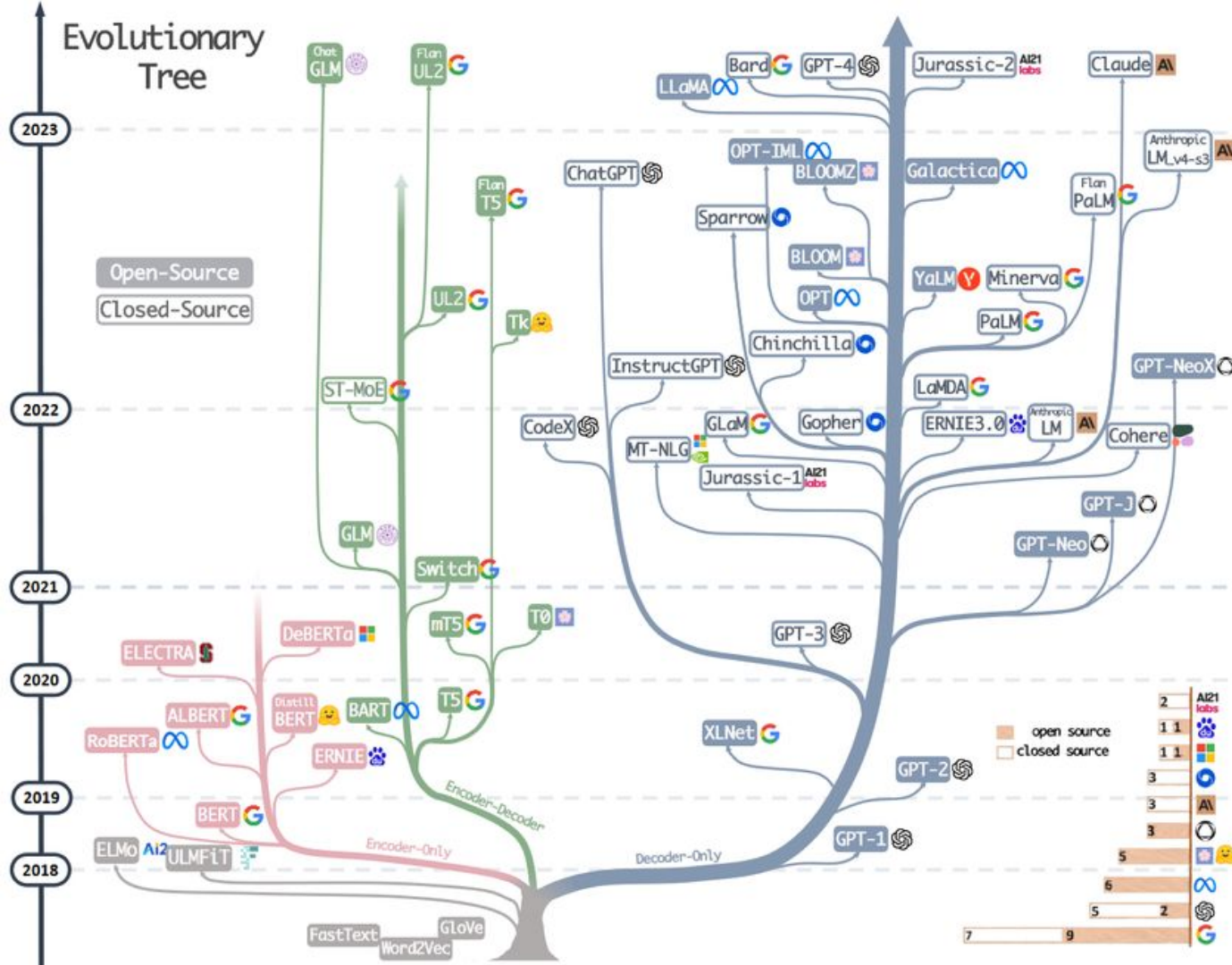
```
def fizz():\n    for i in range(1, 101):\n        if i % 5 == 0 and i % 3 == 0:\n            print("fizzbuzz")\n        elif i % 5 == 0:\n            print("buzz")\n        elif i % 3 == 0:\n            print("fizz")\n        else:\n            print(i)
```



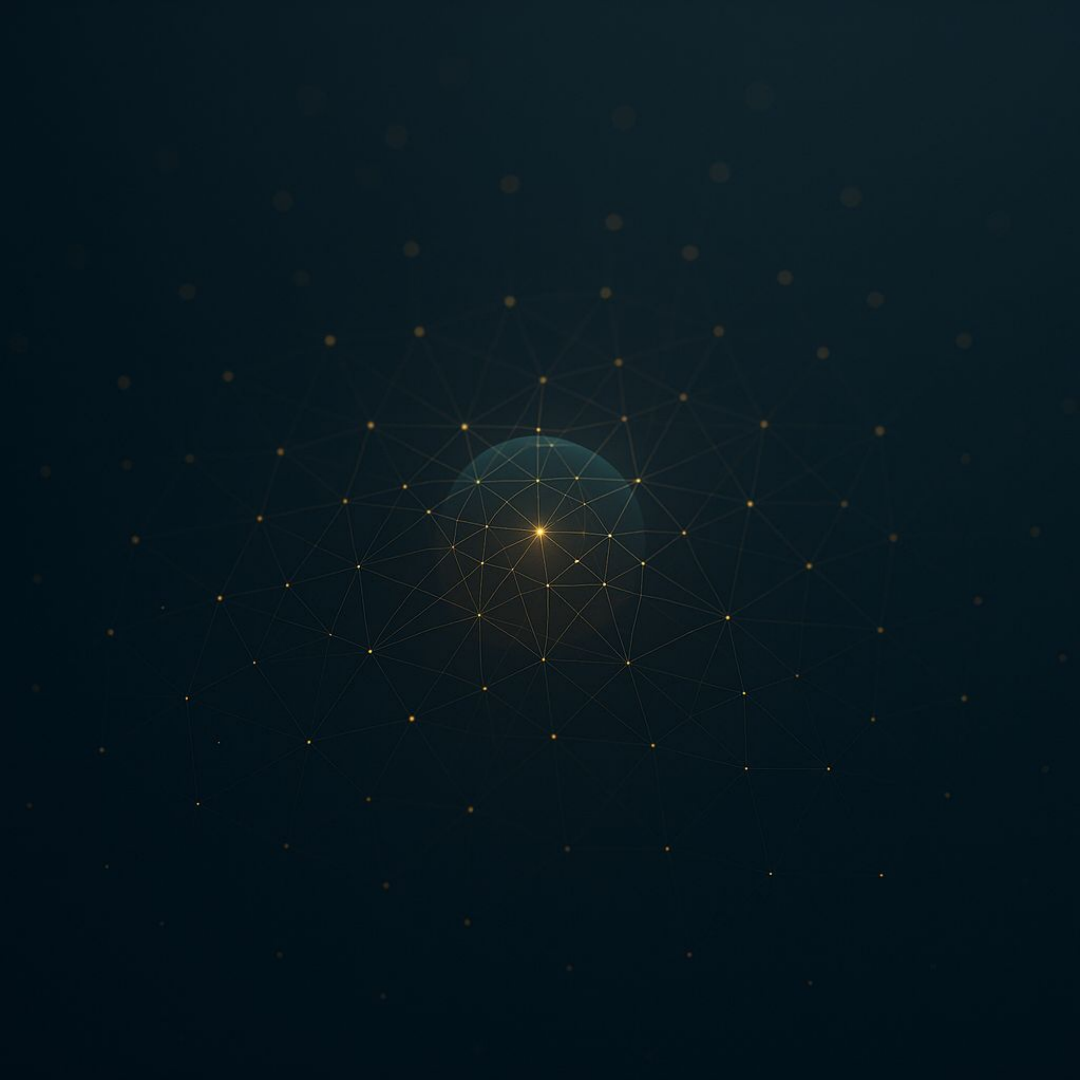
# Transformer

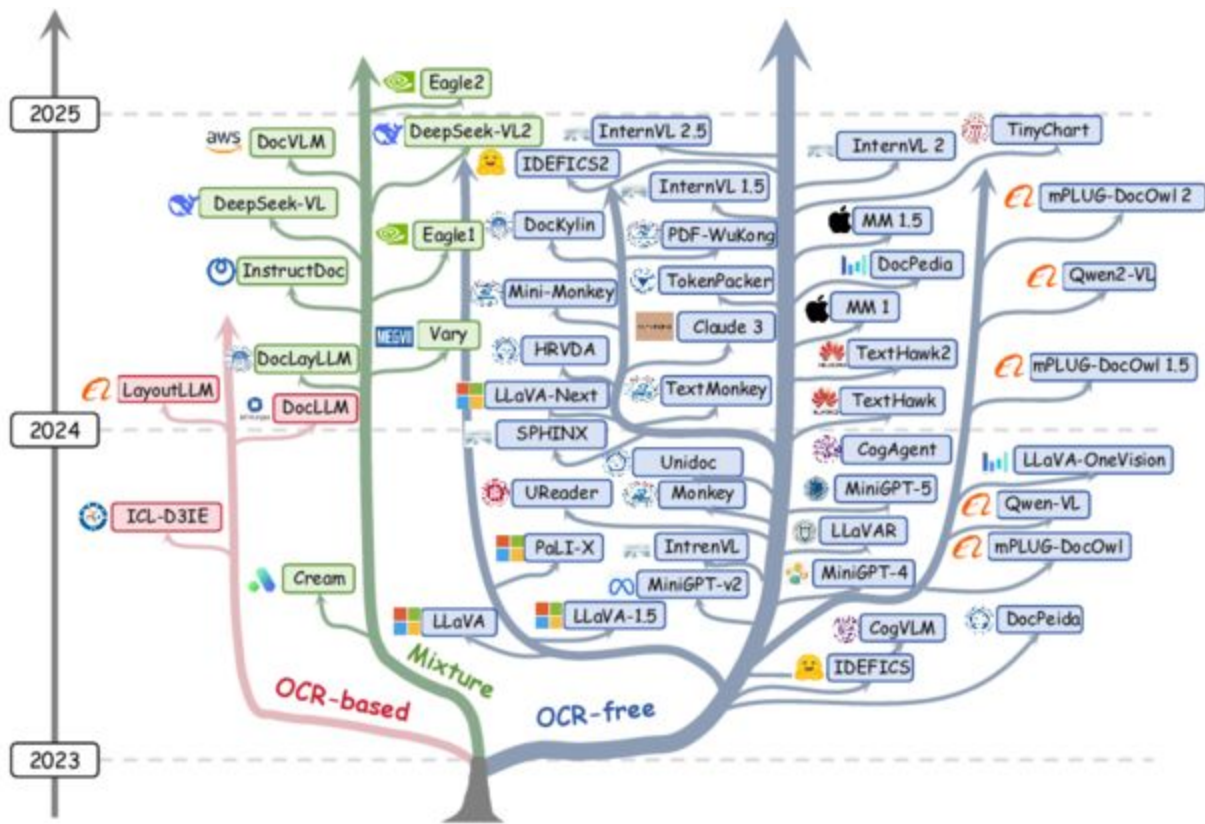
Attention Is All You Need

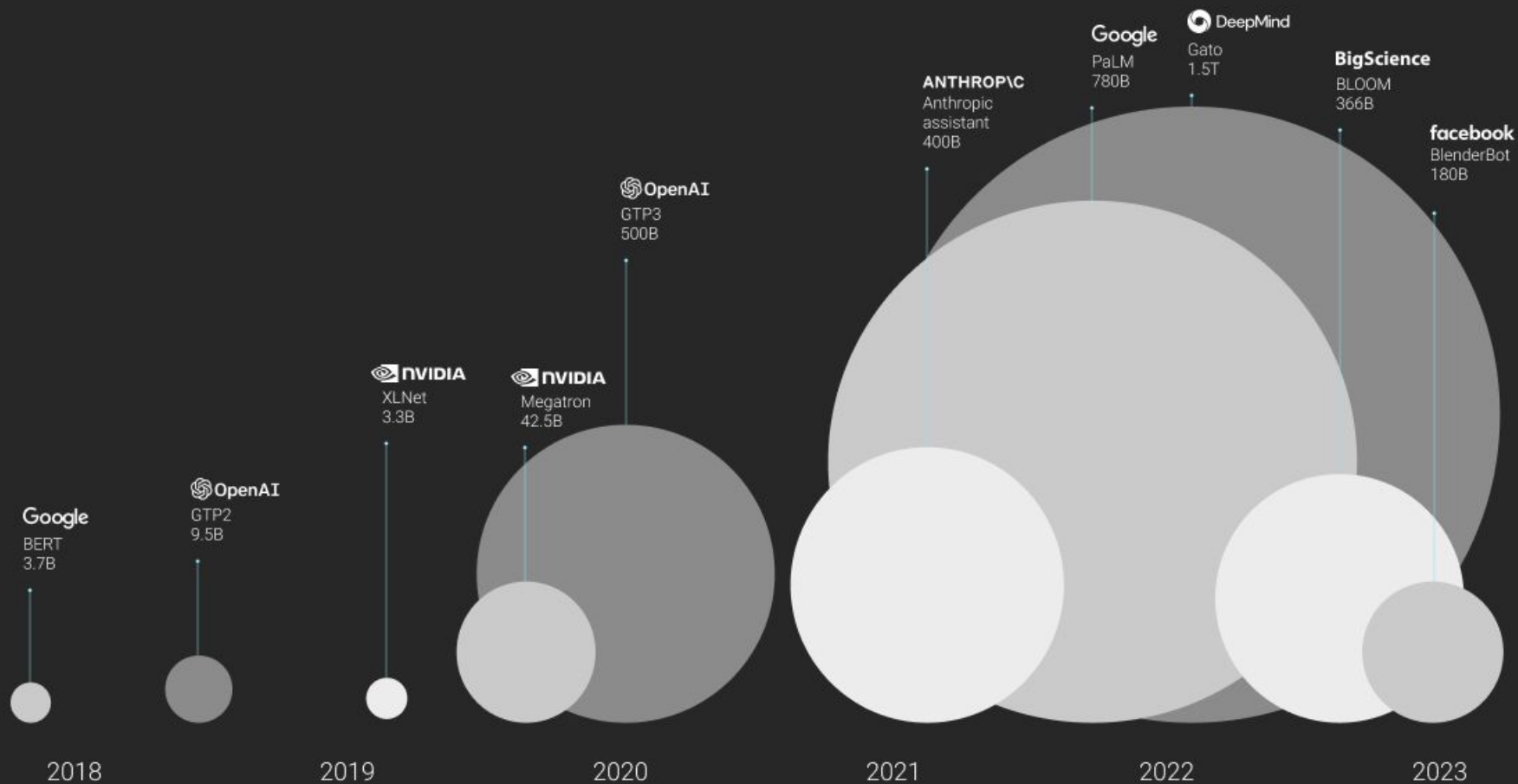




LLM







# SLM: Small (or smart) Language model



r/OpenAI • 2 mo. ago  
bgboy089

...

## GPT-5 is actually a much smaller model

### Discussion

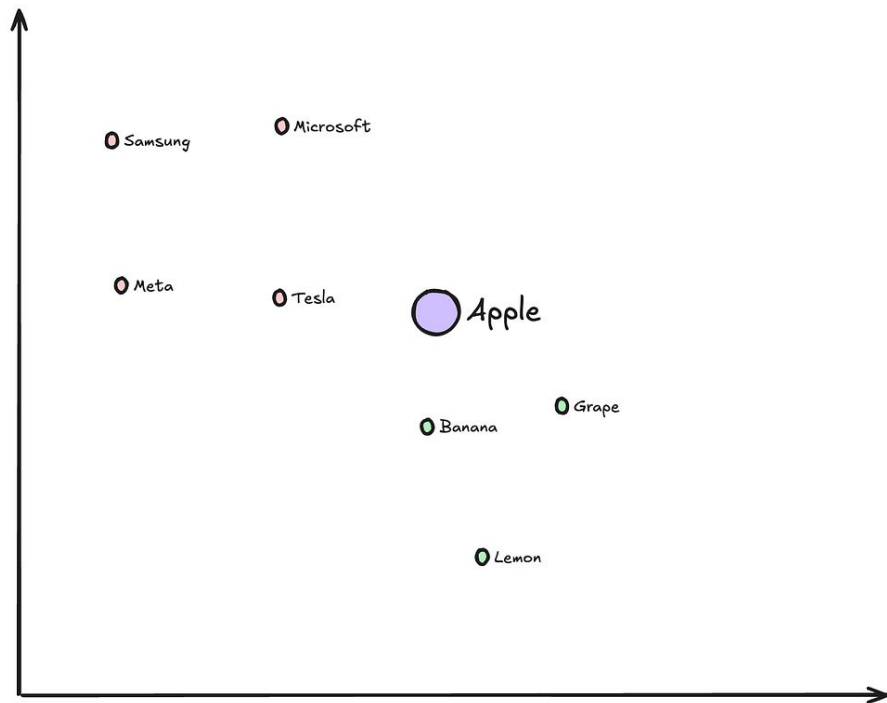
Another sign that GPT-5 is actually a much smaller model: just days ago, OpenAI's O3 model, arguably the best model ever released, was limited to 100 messages per week because they couldn't afford to support higher usage. That's with users paying \$20 a month. Now, after backlash, they've suddenly increased GPT-5's cap from 200 to 3,000 messages per week, something we've only seen with lightweight models like O4 mini.

If GPT-5 were truly the massive model they've been trying to present it as, there's no way OpenAI could afford to give users 3,000 messages when they were struggling to handle just 100 on O3. The economics don't add up. Combined with GPT-5's noticeably faster token output speed, this all strongly suggests GPT-5 is a smaller, likely distilled model, possibly trained on the thinking patterns of O3 or O4, and the knowledge base of 4.5.

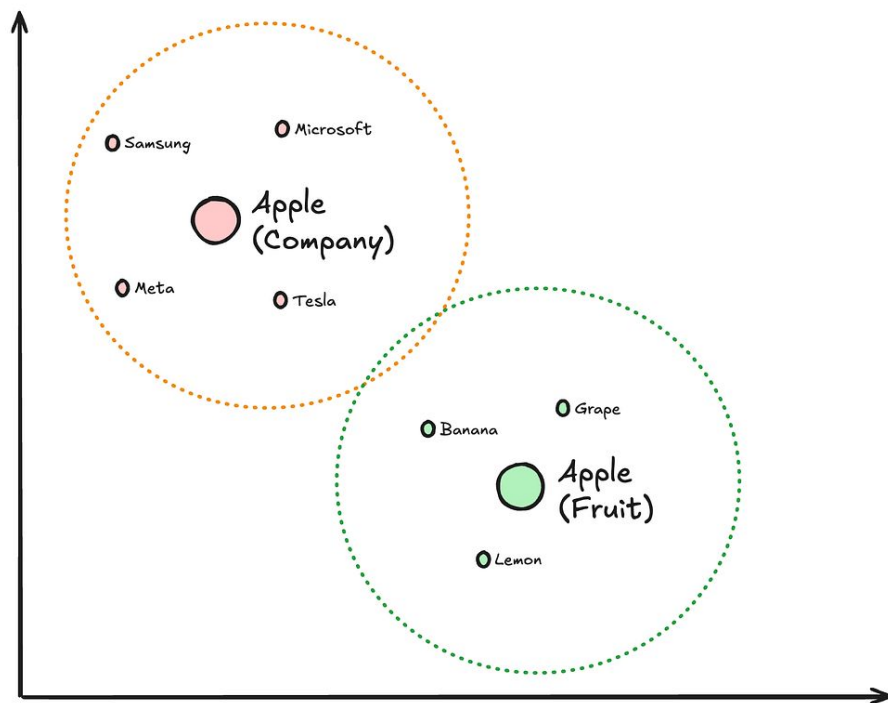
## MoE: Mixture of Experts

Contextual tokens

## Static Embeddings



## Contextual Embeddings





Level 1: input-output



LLM pipelines

RAG



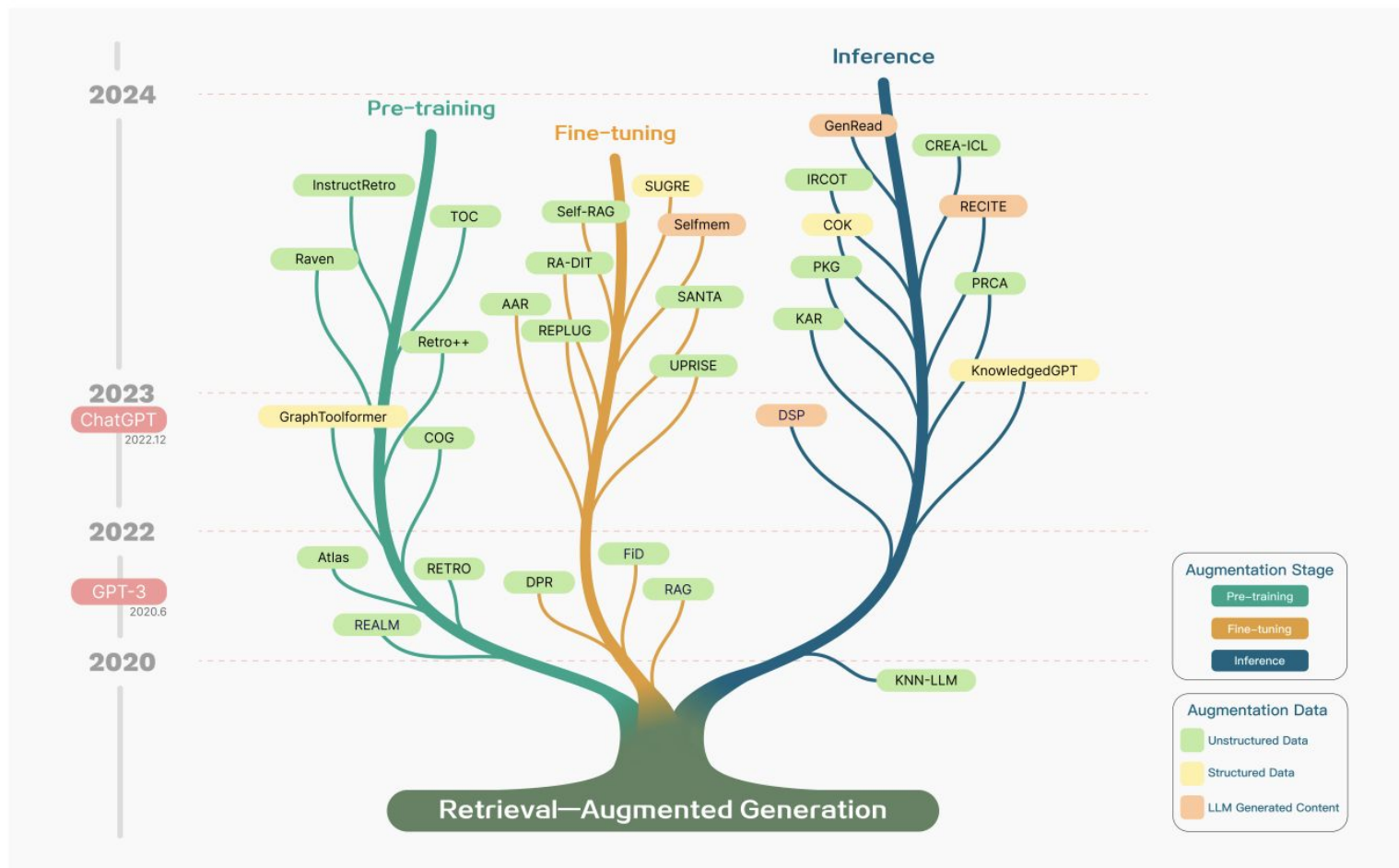
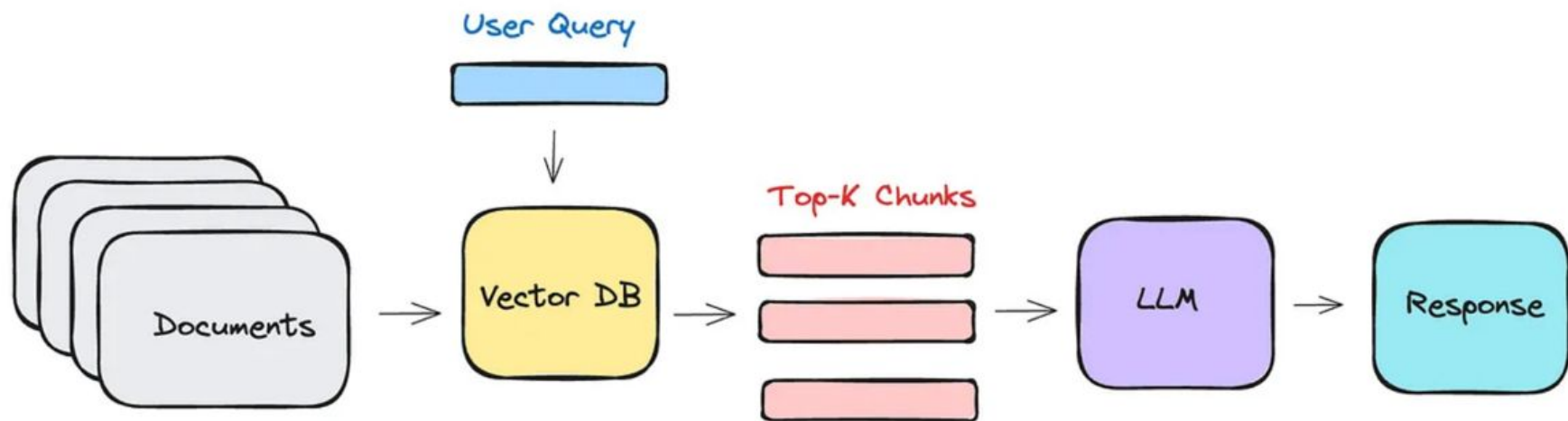


Figure 1: A timeline of existing RAG research. The timeline was established mainly according to the release date.

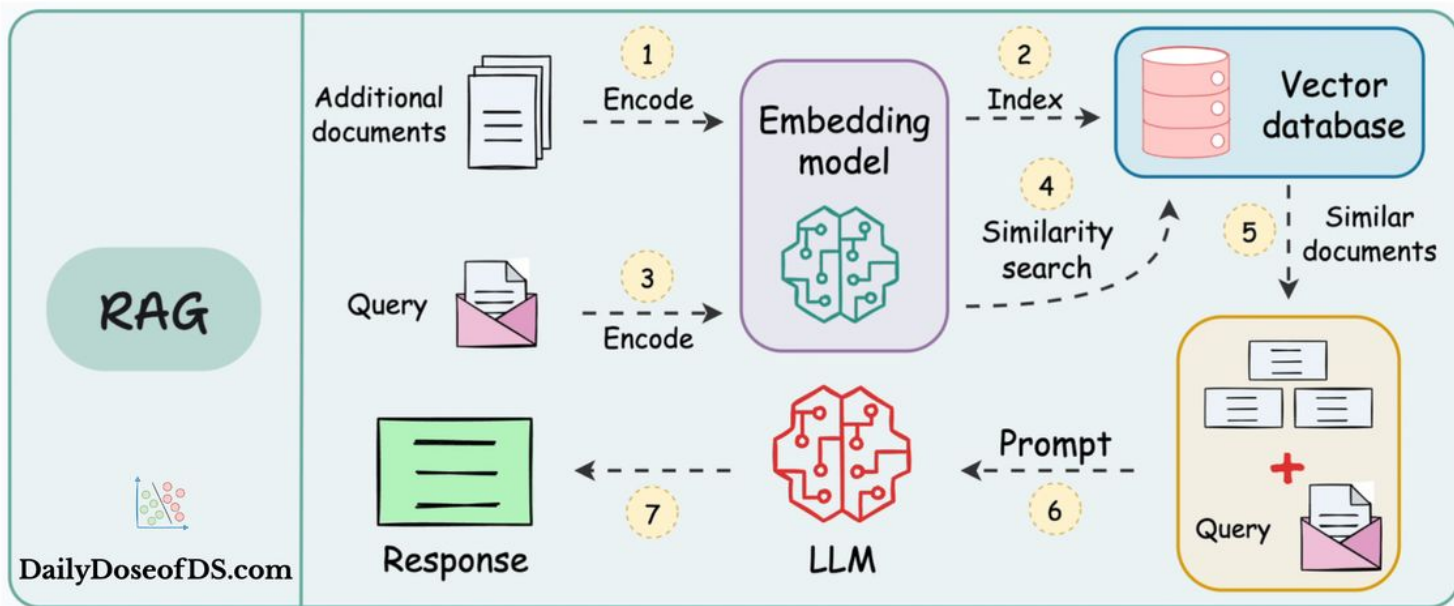
# Basic RAG Pipeline



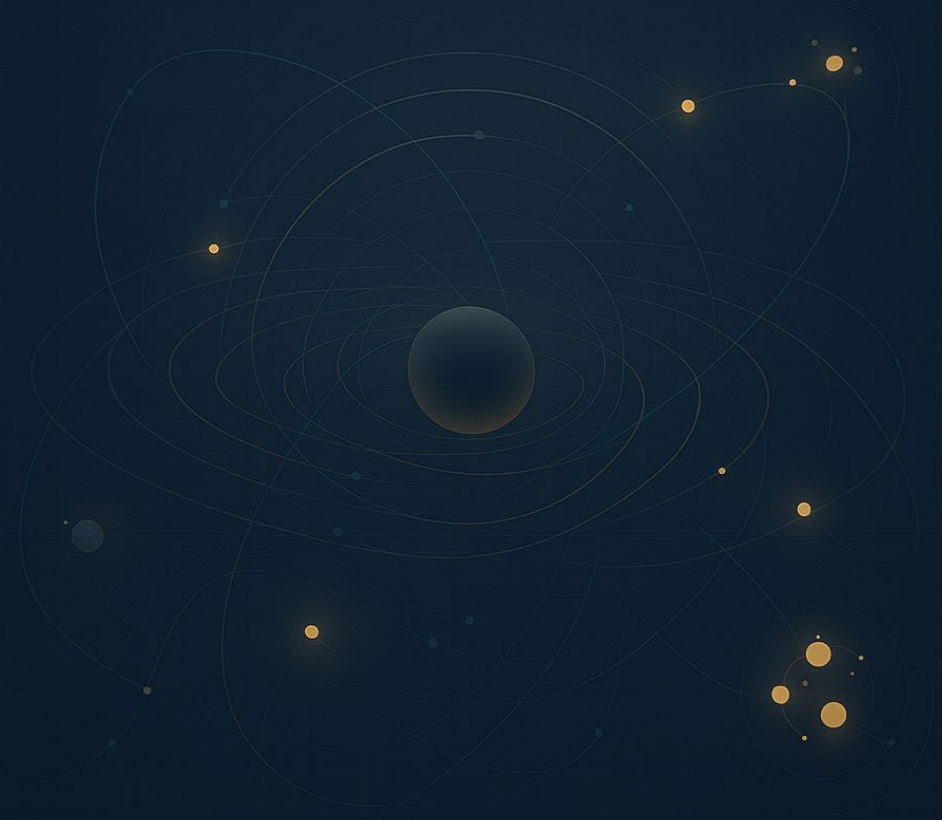
Step 1: Data Indexing

Step 2: Data Retrieval & Generation

# 16 techniques to supercharge RAG systems



# Agentic



**LLM (raw)**

→ get an answer based on the trained model

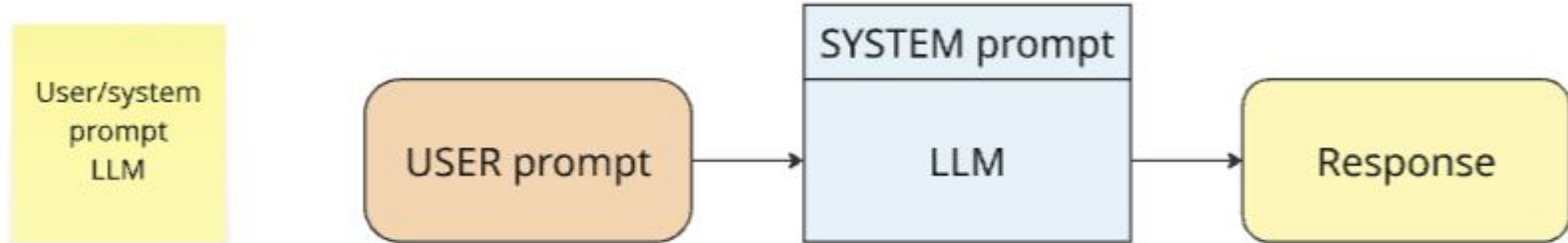
Raw  
LLM



## LLM (system prompt)

→ get an answer based on the trained model

→ Separates request (USER prompt) and model behaviour (POST-training SYSTEM prompt)



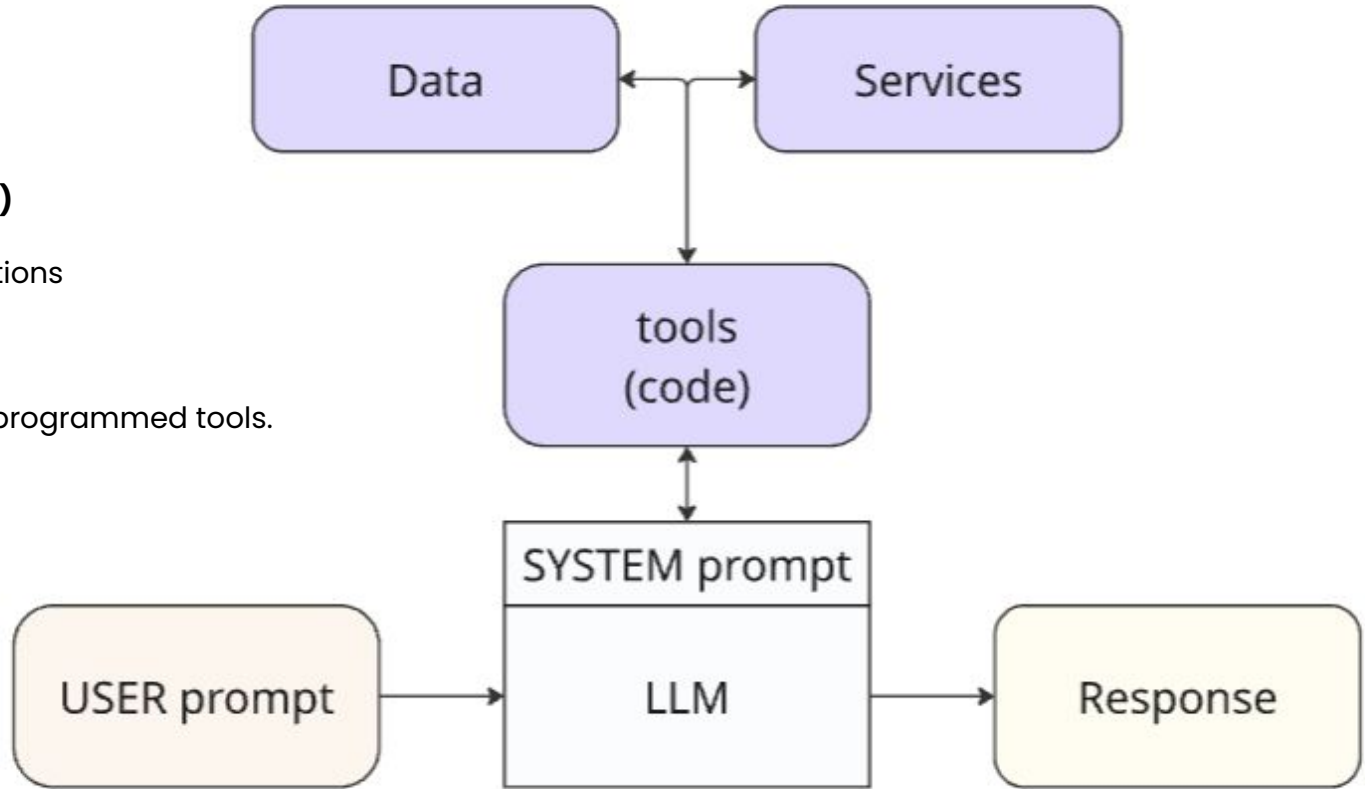
### Tools (functions for LLMS)

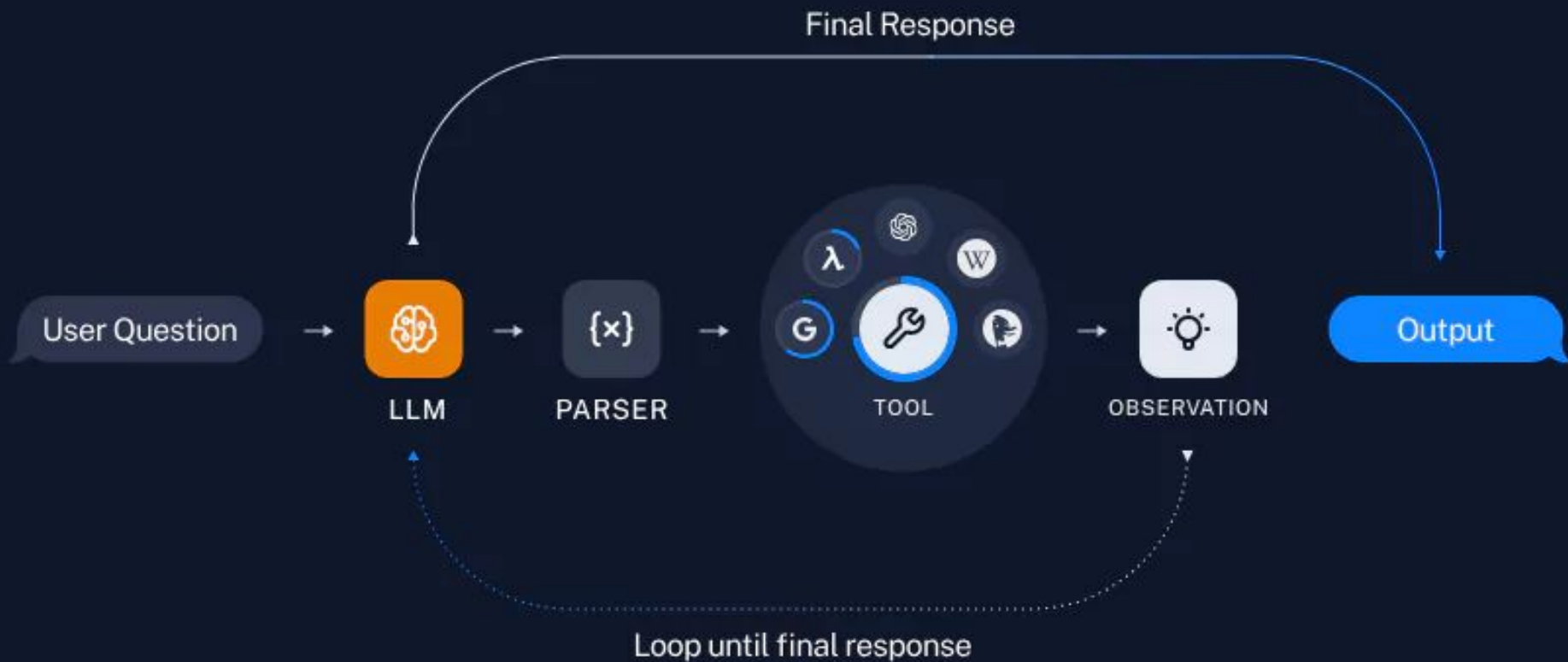
→ Let LLMs run code functions

→ tool-augmented LLM:

Gives LLM access to pre-programmed tools.

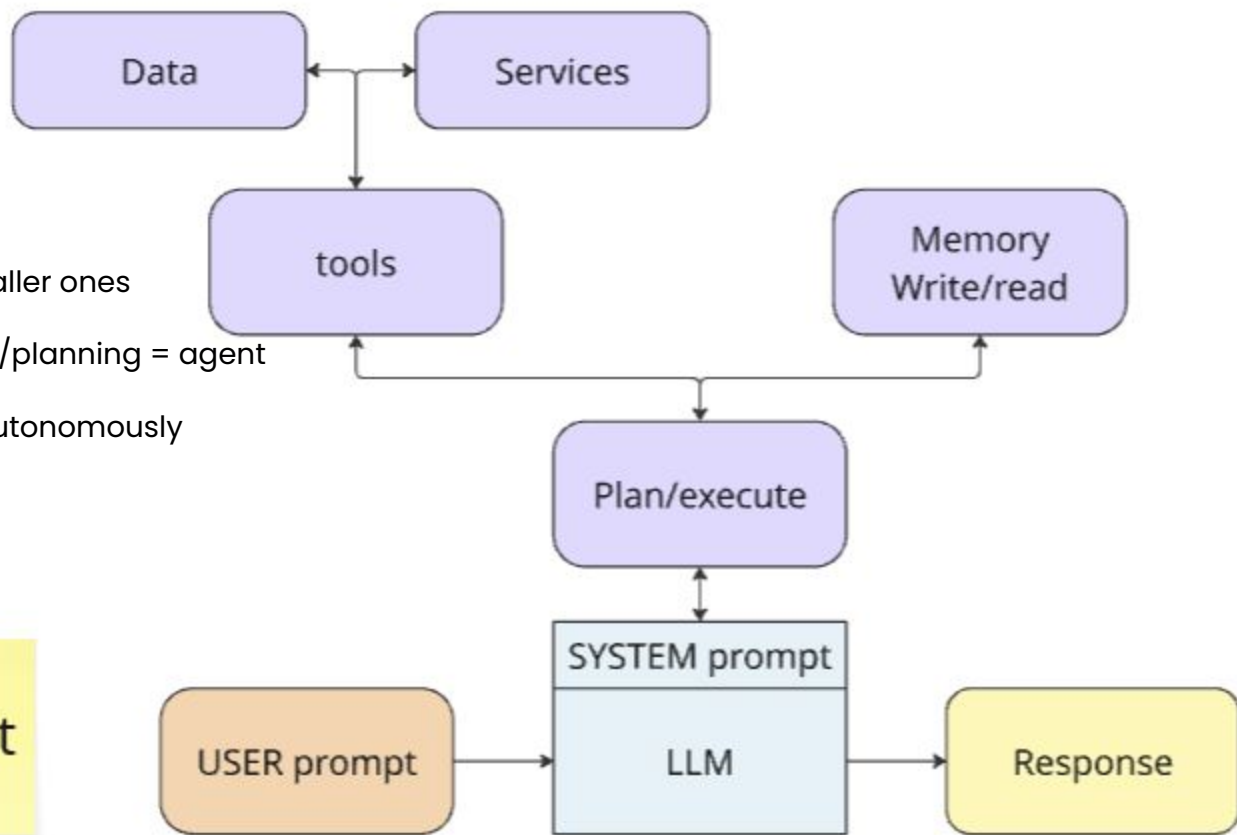
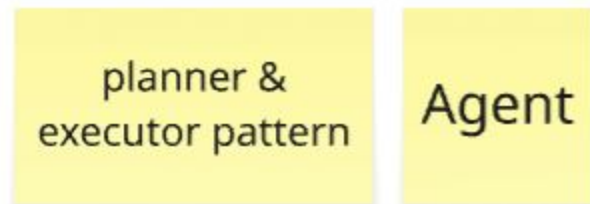
Tools





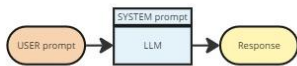
## **Agents** (LLMs with tools & planning)

- Can break down big tasks into smaller ones
- LLM + tools + control loop/memory/planning = agent
- Able to complete requests more autonomously



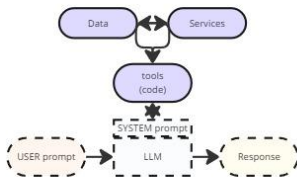


Raw  
LLM



User/system  
prompt  
LLM

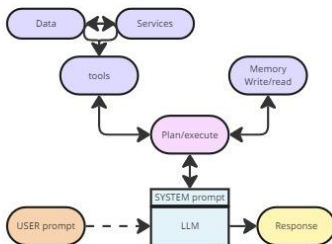
prompt  
engineering



LLM  
+ Tools

prompt  
engineering

Tools



Agent

Context  
engineering

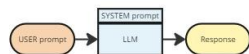
Tools  
Subagents

reasoning

MCP

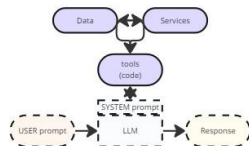


Raw  
LLM



User/system  
prompt  
LLM

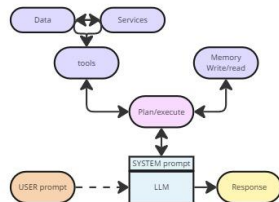
prompt  
engineering



LLM  
+ Tools

prompt  
engineering

Tools



Agent

Context  
engineering

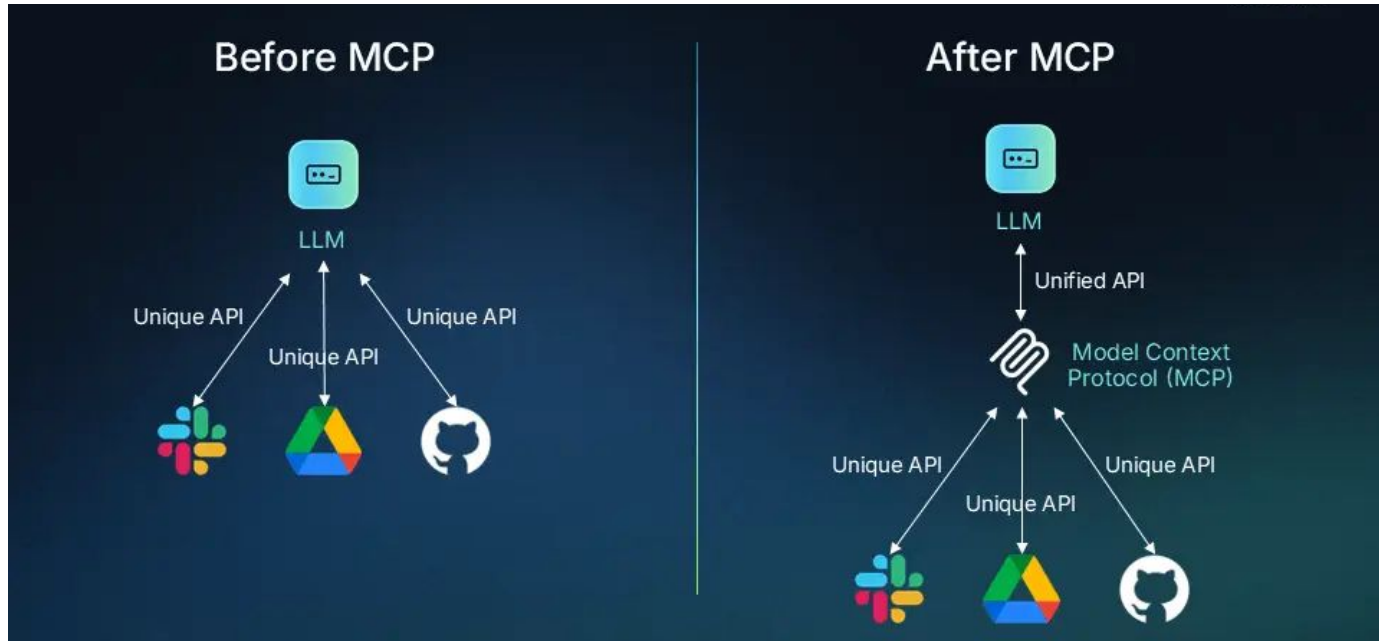
Tools  
Subagents

reasoning

MCP

## MCP (Model Context Protocol)

- standardized interface for how models access tools, prompts, and resources.
- equip local LLM agent with discoverable external tools.



# Server Key Components

MODEL CONTROLLED

## Tools

Functions invoked by the model

Retrieve / search

Send a message

Update DB records

APPLICATION CONTROLLED

## Resources

Data exposed to the application

Files

Database Records

API Responses

USER CONTROLLED

## Prompts

Pre-defined templates for AI interactions

Document Q&A

Transcript Summary

Workflow Automation

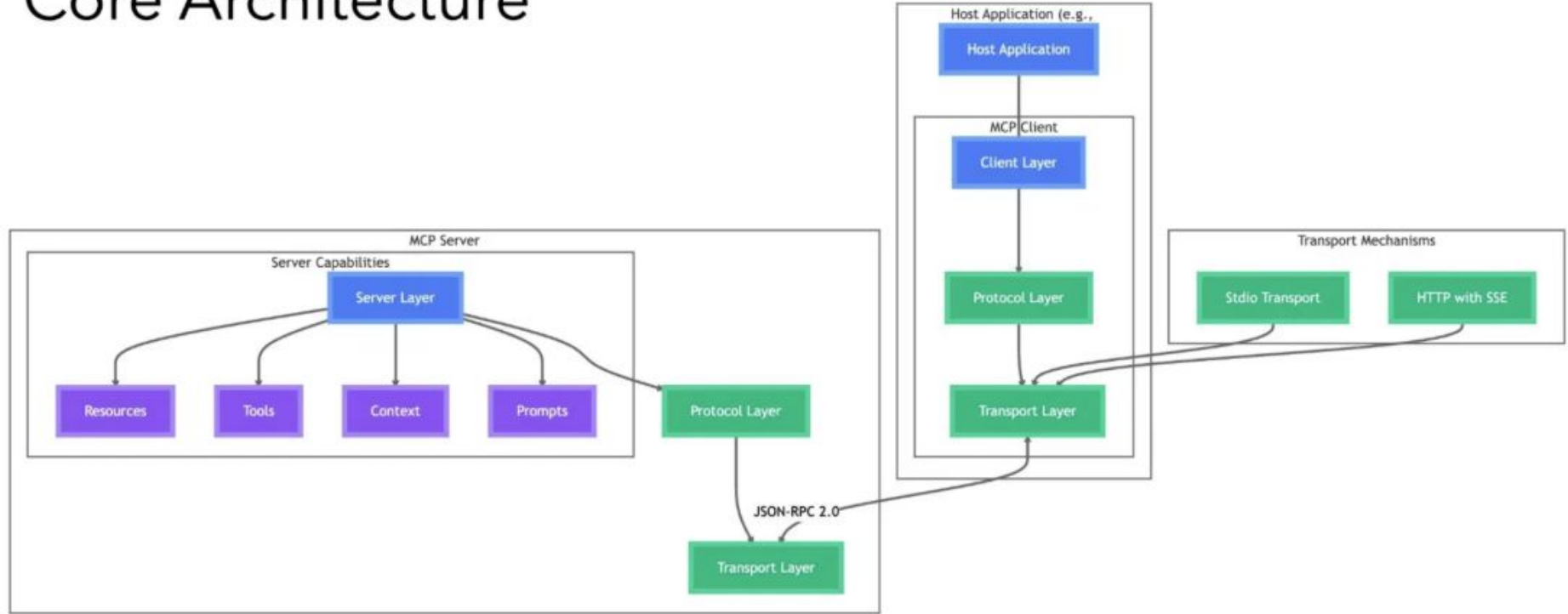
## Protocol Layer

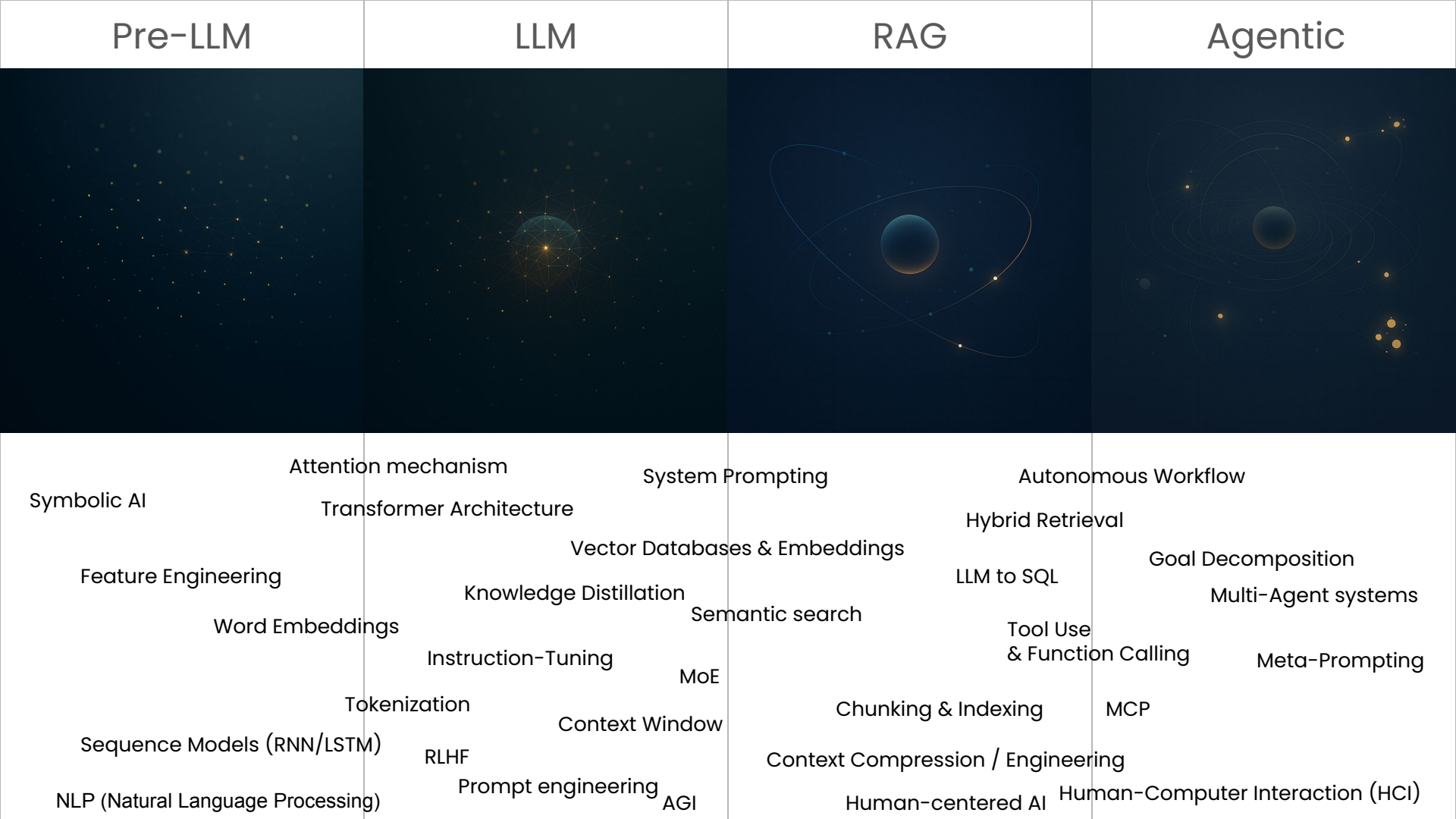
Message Framing & Routing  
Request/Response Management  
Communication Patterns  
Versioned Protocol Schemas

## Transport Layer

JSON-RPC 2.0 Wire Format  
Protocol Message Transmission  
Multiple Transport Methods  
Extensible Channel System

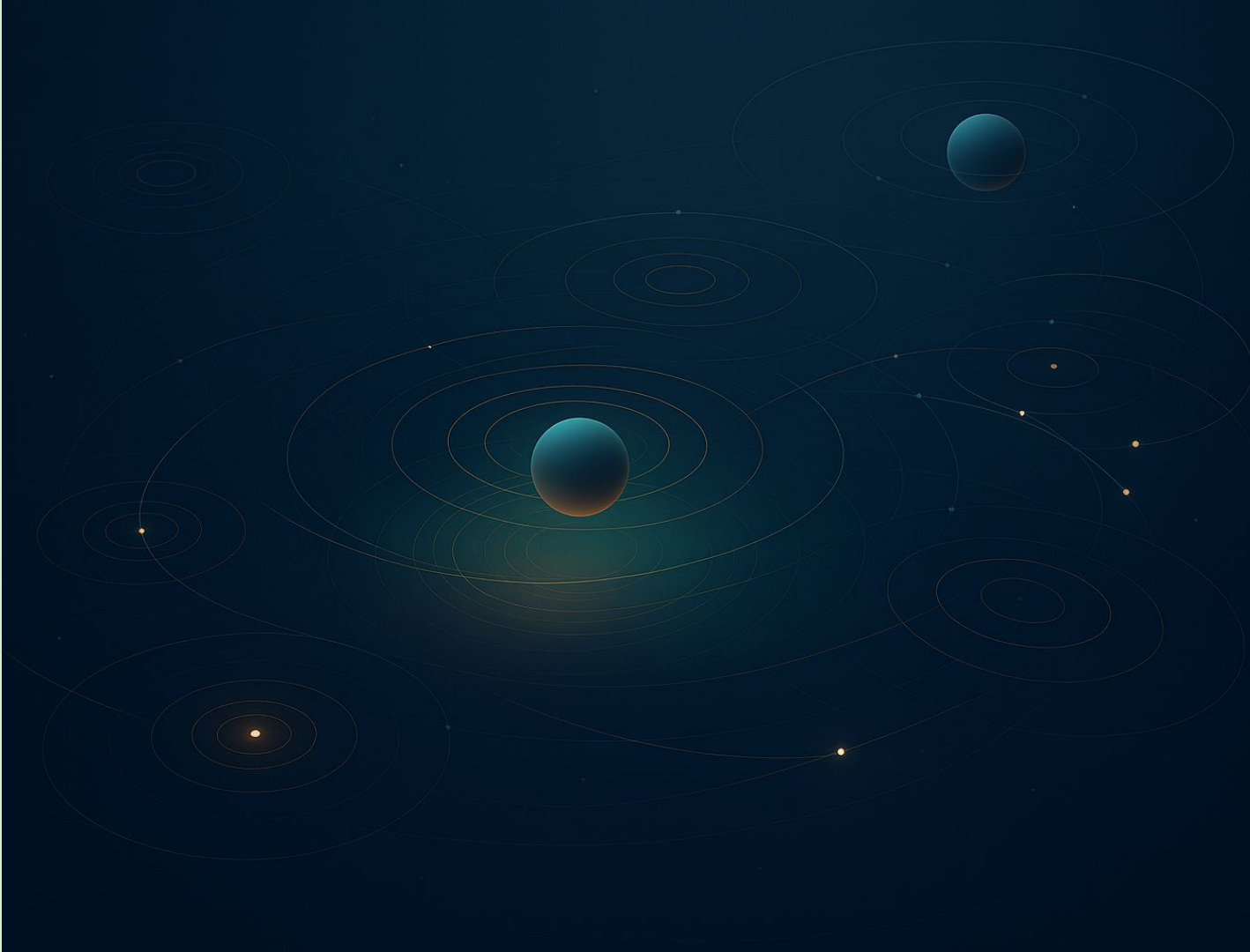
# Core Architecture





Whats next?

# Agentic economy



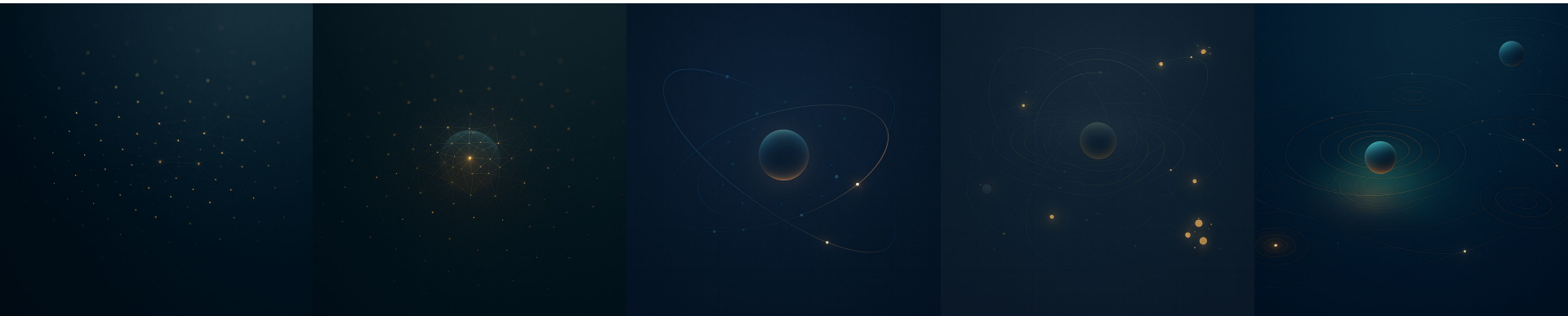
Pre-LLM

LLM

RAG

Agentic

Agentic economy



Data-pipelines  
Model-pipelines  
  
Vectors &  
embeddings

LLMs  
  
(as a  
data-pipeline)

RAG  
  
(as a  
LLM-pipeline)

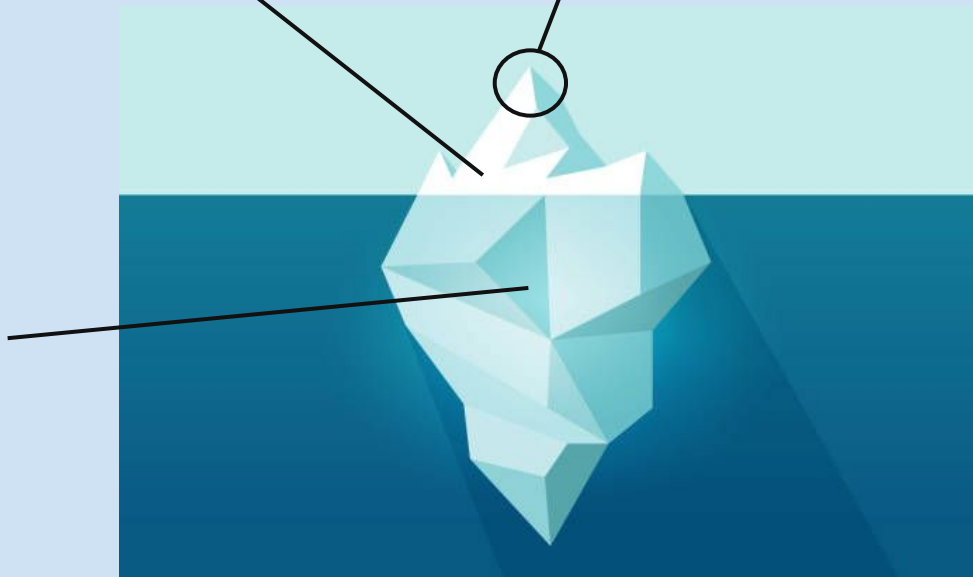
Agent  
  
(as an LLM with  
reasoning and  
tools)

Agent to Agent  
  
An economy of  
agents,  
exchanging  
information

# Techichal

Societal,  
mental/cognitive,  
political,  
Economic,  
Cultural,  
Academic,  
Environmental,  
Ethical,  
Educational

This presentation



“Crash course in LLMs”

End / Q&a

**Prompt Engineering Guide** - <https://www.promptingguide.ai/>

**Daily Dose of Data Science** - <https://www.dailydoseofds.com/>

**Model Context Protocol** - <https://modelcontextprotocol.io/>

Costs

Effects

Costs

Effects

